Server Migration Service

User Guide

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HUAWEI CLOUD COMPUTING TECHNOLOGIES CO., LTD.

Contents

1 Permissions Management

1.1 Using IAM to Grant Access to SMS

1.1.1 Using IAM Roles or Policies to Grant Access to SMS

System-defined permissions **in role/policy-based authorization** provided by **IAM** let you control access to SMS. With IAM, you can:

- Create IAM users for employees based on the organizational structure of your enterprise. Each IAM user has their own security credentials, providing access to SMS resources.
- Grant users only the permissions required to perform a given task based on their job responsibilities.
- Entrust an account or cloud service to perform professional and efficient O&M on your SMS resources.

If your account does not need individual IAM users for permissions management, you can skip this section.

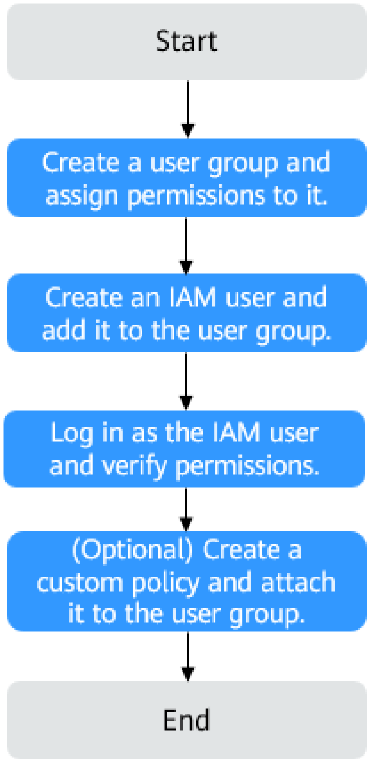
Figure 1-1 shows the process flow of role/policy-based authorization.

Prerequisites

Before granting permissions to user groups, learn about **Role/Policy-based Authorization** for SMS. To grant permissions for other services, learn about all **system-defined permissions**supported by IAM.

Process Flow

Figure 1-1 Process for granting SMS permissions



1. **Create a user group and assign it permissions**.

   You can assign system-defined policies and custom policies to the user group. Custom policies extend and supplement system-defined policies, providing more refined access control. Select proper authorization policies based on service requirements.

   – If the IAM users in the user group need all permissions for SMS, attach system-defined policies to the user group. On the IAM console, create a user group and attach the SMS FullAccess, OBS OperateAccess, ECS FullAccess, VPC FullAccess, IMS FullAccess, EVS FullAccess, and EIP FullAccess policies to the user group. If disk encryption is required, EVS KMSAccess must also be attached.

   – If the IAM users only need specific SMS permissions, create custom policies and attach these policies to the user group. For details about the content and creation method of custom policies, see **Example Custom Policies**.

2. **Create an IAM user and add it to the created user group**.

   Create an IAM user and add it to the user group created in **1**.

3. **Log in as the IAM user** and verify permissions.

   In the authorized region, perform the following operations:

   – Choose **Service List** > **Server Migration Service**. In the navigation pane on the left, click **Servers**. In the server list, locate the server to be migrated, and click **Configure** in the **Target** column to configure the target server. If the target server can be configured, the permissions have taken effect.

– Choose a service other than SMS and its dependents services in the **Service List**. If a message appears indicating that you have insufficient permissions to access the service, the permissions have taken effect.

Example Custom Policies

1. Custom policies can be created to supplement the system-defined policies of SMS.

   To create a custom policy, choose either visual editor or JSON.

   – Visual editor: Select cloud services, actions, resources, and request conditions. This does not require knowledge of policy syntax.

   – JSON: Create a JSON policy or edit an existing one.

   For details, see **Creating a Custom Policy**. The following lists examples of common SMS custom policies.

- Example SMS policy that contains permissions for project-level services

```
{
    "Version": "1.1",
    "Statement": [
        {
            "Action": [
                "vpc:securityGroups:create",
                "vpc:securityGroupRules:create",
                "vpc:vpcs:create",
                "vpc:publicIps:create",
                "vpc:subnets:create",
                "ecs:cloudServers:create",
                "ecs:cloudServers:attach",
                "ecs:cloudServers:detachVolume",
                "ecs:cloudServers:start",
                "ecs:cloudServers:stop",
                "ecs:cloudServers:delete",
                "ecs:cloudServers:reboot",
                "ecs:cloudServers:updateMetadata",
                "ecs:serverPasswords:manage",
                "ecs:serverKeypairs:delete",
                "ecs:diskConfigs:use",
                "ecs:CloudServers:create",
                "ecs:servers:setMetadata",
                "ecs:serverVolumes:use",
                "ecs:serverKeypairs:create",
                "ecs:serverInterfaces:use",
                "ecs:serverGroups:manage",
                "ecs:securityGroups:use",
                "ecs:servers:unlock",
                "ecs:servers:rebuild",
                "ecs:servers:lock",
                "ecs:servers:reboot",
                "evs:volumes:use",
                "evs:volumes:create",
                "evs:volumes:update",
                "evs:volumes:delete",
                "evs:volumes:attach",
                "evs:volumes:detach",
                "evs:snapshots:create",
                "evs:snapshots:delete",
                "evs:snapshots:rollback",
                "kms:cmk:list",
                "kms:cmk:get",
                "kms:dek:create",
                "kms:dek:decrypt",
```



```
                "ecs:*:get*",
                "ecs:*:list*",
                "evs:*:get*",
                "evs:*:list*",
                "vpc:*:list*",
                "vpc:*:get*",
                "ims:*:get*",
                "ims:*:list*"
            ],
            "Effect": "Allow"
        }
    ]
}
```

- Example SMS policy that contains permissions for global services

```
{
    "Version": "1.1",
    "Statement": [
        {
            "Effect": "Allow",
            "Action": [
                "sms:server:registerServer",
                "sms:server:migrationServer",
                "sms:server:queryServer",
                "iam:roles:listRoles",
                "iam:agencies:listAgencies",
                "iam:permissions:listRolesForAgency"
            ]
        }
    ]
}
```

1.1.2 Using IAM Identity Policies to Grant Access to SMS

System-defined permissions **in identity policy-based authorization** provided by **IAM** let you control access to SMS. With IAM, you can:

- Create IAM users or user groups for personnel based on your enterprise's organizational structure. Each IAM user has their own identity credentials for accessing SMS resources.
- Grant users only the permissions required to perform a given task based on their job responsibilities.
- Entrust a Huawei Cloud account or a cloud service to perform efficient O&M on your SMS resources.

If your Huawei Cloud account meets your permissions requirements, you can skip this section.

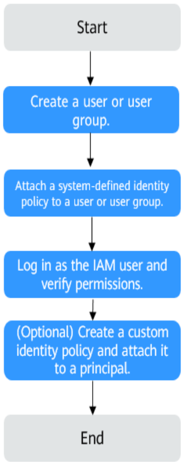
Figure 1-2 shows the process flow of identity policy-based authorization.

Prerequisites

Before granting permissions, learn about **Identity Policy-based Authorization** for SMS. To grant permissions for other services, learn about all **system-defined permissions** supported by IAM.

Process Flow

Figure 1-2 Process for granting SMS permissions



1. On the IAM console, **create an IAM user** or **create a user group**.
2. **Attach a system-defined policy** (SMSReadOnlyPolicy as an example) to the user or user group.
3. **Log in as the IAM user** and verify permissions.

   In the authorized region, perform the following operations:

   – Choose **Service List** > **Server Migration Service**. On the SMS console, locate the server to be migrated and click **Configure** in the **Target** column. If a message appears indicating insufficient permissions to perform the operation, the **SMSReadOnlyPolicy** policy is in effect.

   – Choose any other service in the **Service List**. If a message appears indicating insufficient permissions to access the service, the **SMSReadOnlyPolicy** policy is in effect.

Example Custom Policies

You can create custom identity policies to supplement the system-defined identity policies of SMS.

To create a custom identity policy, choose either visual editor or JSON.

- Visual editor: Select cloud services, actions, resources, and request conditions. This does not require knowledge of policy syntax.
- JSON: Create a JSON policy or edit an existing one.

For details, see **Creating a Custom Identity Policy and Attaching It to a Principal**.

When creating a custom Identity policy, use the Resource element to specify the resources the policy applies to and use the Condition element (service-specific condition keys) to control when the policy is in effect. The following is an example of custom identity policies for SMS.

- Example 1: Grant permissions to query server details.

```
{
    "Version": "5.0",
    "Statement": [
        {
            "Effect": "Allow",
            "Action": [
                "sms:server:get"
            ]
        }
    ]
}
```

- Example 2: Grant permissions to list, query, and delete migration tasks.

```
{
    "Version": "5.0",
    "Statement": [
        {
            "Effect": "Allow",
            "Action": [
                "sms:server:deleteTask",
                "sms:server:getTask",
                "sms:server:listTask"
            ]
        }
    ]
}
```

2 Installing the Agent on the Source Server

2.1 Installing the SMS-Agent on Windows

Without installing the Agent on the source server, SMS cannot collect server information or initiate a migration task. Before the migration, you must install the Agent on the source server to be migrated. During the installation, you need to enter the AK/SK pair of the Huawei Cloud account you are migrating to. After the Agent is started, it automatically reports source server information to SMS. The information is used for migration only. To learn what source server information is collected by SMS, see **What Information Does SMS Collect About Source Servers?**

NOTE

Before using SMS to migrate servers, you need to manually install and register the SMS-Agent on each server to be migrated. If there are more than 50 servers to migrate, you can **create a batch server migration workflow** on MgC to automate batch installation and registration of the Agent.

The SMS-Agent version varies depending on the Windows version. For details, see **Table 2-1**.

Table 2-1 SMS-Agent versions

<table>
<tr><th>OS Version</th><th>SMS-Agent version</th><th>Description</th></tr>
<tr><td>Windows Server 2012, Windows Server 2016, Windows Server 2019, Windows Server 2022, Windows Server 2025, Windows 8.1, Windows 10, and Windows 11</td><td>GUI-based Windows Agent (Python 3)</td><td rowspan="2">Windows Agent (Python 2) can run on Windows of later versions.</td></tr>
<tr><td>Windows Server 2008 and Windows 7</td><td>CLI-based Windows Agent (Python 2)</td></tr>
</table>

Constraints

You need to log in to the source Windows server as Administrator.

Prerequisites

- You have obtained an AK/SK pair for your Huawei Cloud account.
  - If you use an IAM user for migration, obtain an AK/SK pair by referring to **How Do I Obtain an AK/SK Pair for an IAM User?**
  - If you use an account for migration, obtain an AK/SK pair by referring to **How Do I Obtain an AK/SK Pair for a Huawei Cloud Account?**
- You have obtained the administrator permissions for the source server.
- You have confirmed that the source server OS is supported by SMS. Learn more about **supported Windows OSs**.
- There is no antivirus software on the source server. Antivirus software may prevent the Agent from starting up.

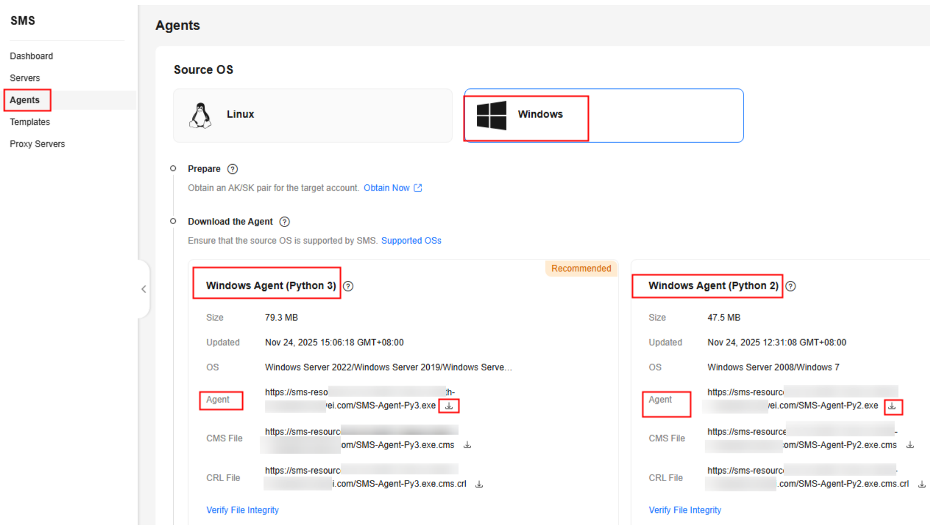
Downloading the SMS-Agent Installation File

Step 1 Sign in to the **SMS console**.

Step 2 In the navigation pane, choose **Agents**. The **Agents** page is displayed.

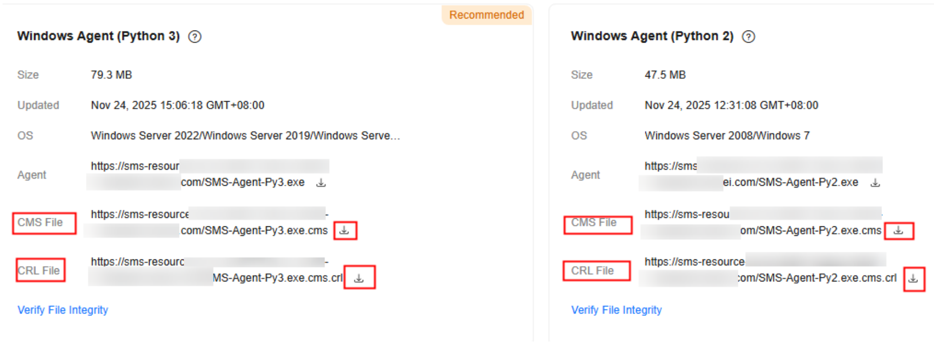
Step 3 Select the **Windows** card, locate the Agent that matches the source server OS, and click the icon next to **Agent**.

- Windows Agent (Python 3): Windows Server 2012, Windows Server 2016, Windows Server 2019, Windows Server 2022, Windows Server 2025, Windows 8.1, Windows 10, and Windows 11
- CLI-based Windows Agent (Python 2): Windows Server 2008 and Windows 7



Step 4 Read and agree to the service agreement, and click **Yes** to download the Agent installation file.

Step 5 Click next to **CMS File** and **CRL File** to download them to the source server.



Step 6 Verify the integrity of the Agent installation file. For details, see **How Do I Verify the Integrity of the Agent Installation File?**

----End

Installing the SMS-Agent

Select the installation method based on the OS version.

- Windows Agent (Python 3): Windows Server 2012, Windows Server 2016, Windows Server 2019, Windows Server 2022, Windows Server 2025, Windows 8.1, Windows 10, and Windows 11
- Windows Agent (Python 2): Windows Server 2008 and Windows 7

Installing the Windows Agent (Python 3)

Step 1 Transmit the **SMS-Agent-Py3.exe** file to the source server.

Step 2 Log in to the source server as user **Administrator** and double-click the **SMS-Agent-Py3.exe** file.

Step 3 Click **Install** and wait for the installation to complete.

Step 4 Click **Finish** to open the SMS-Agent GUI.

Step 5 Enter the AK/SK pair for the Huawei Cloud account that you are migrating to and the SMS domain name. You can obtain the SMS domain name on the **Agents** page of the SMS console, as shown in **Figure 2-2**.

Figure 2-1 Starting the Agent

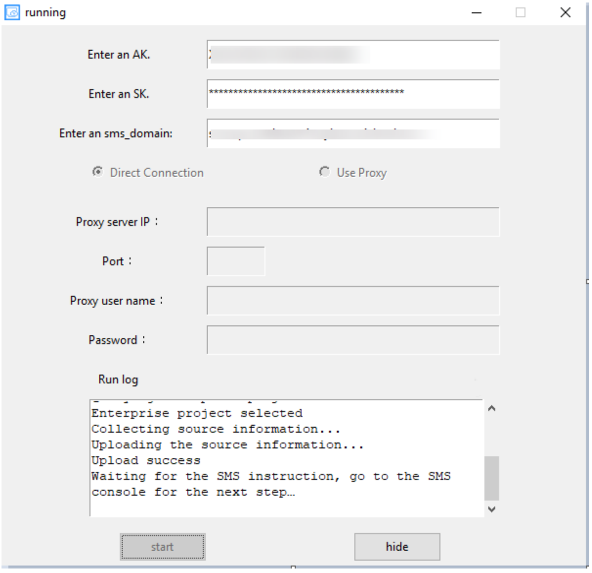
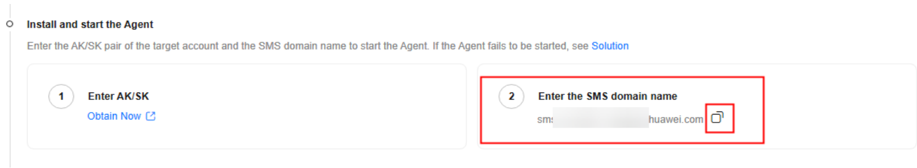


Figure 2-2 Obtaining the SMS domain name



Step 6 Determine whether to use an HTTP/HTTPS proxy server for the migration.

- If you do not need to use an HTTP/HTTPS proxy, select **Direct Connection**.
- If you need to use an HTTP/HTTPS proxy, select **Use Proxy** and set parameters such as the IP address, port number, username, and password of the proxy server based on **Table 2-2**.

Table 2-2 Proxy parameter settings

Parameter	Description
Proxy server IP	Enter the IP address of the proxy server, not that of the target server. The input format is https://your-proxy-addr.com. Replace *your-proxy-addr.com* with the actual proxy server address. HTTPS is recommended.
Port	Enter the proxy port opened on the proxy server.
External user name	Enter a proxy username if any; otherwise, leave it blank.
Password	Enter a proxy password if any; otherwise, leave it blank.

Step 7 If the EPS service has been enabled for your Huawei Cloud account, after you entered the AK/SK pair, the Agent will list all enterprise projects your account is allowed to access. You can select the enterprise project you would like to migrate the source server to. This enables you to isolate permissions, resources, and finance during the migration. For details, see **Migrating a Server into an Enterprise Project**.

Step 8 Click **start**.

Step 9 Carefully review the **Privacy Statement** and click **Yes** if you want to continue.

When the message "Upload success. Waiting for the SMS instruction" is displayed, the SMS-Agent has been started. You can sign in to the SMS console and perform subsequent operations.

----End

Installing the Windows Agent (Python 2)

Step 1 Transmit the **SMS-Agent-Py2.exe** file to the source server.

Step 2 Log in to the source server as user **Administrator** and double-click the **SMS-Agent-Py2.exe** file.

Step 3 Click **Install** and wait for the installation to complete.

Step 4 Click **Finish** to open the SMS-Agent CLI.

> NOTE
>
> If you need to rerun the SMS-Agent, double-click **agent-start.exe** in the **C:\SMS-Agent-Py2** directory.

Step 5 Determine whether to use an HTTP/HTTPS proxy server for the migration. If your source server cannot access Huawei Cloud over the Internet, you can use a proxy server. You will need to configure the proxy server yourself. In a migration over a private line or VPN, a proxy server is only used for registering the source server with SMS. It is not used for data migration.

- If you do not need an HTTP/HTTPS proxy server, skip this step and go to **the next step**.
- If you need to use an HTTP/HTTPS proxy, do as follows:
  a. Go to the directory where the Agent was installed (typically **C:\SMS-Agent-Py2\config**).
  b. Edit the **auth.cfg** file. Do not edit the **auth.cfg** file unless you need to use an HTTP/HTTPS proxy.

```
[proxy-config]
enable = true
proxy_addr = https://<your-proxy-address>.com
proxy_port = <proxy-port>
proxy_user =
use_password = false
```

Configure the parameters according to the table below.

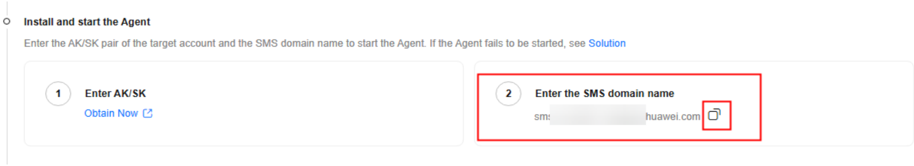
Table 2-3 Proxy parameter settings

Parameter	Description
enable	To use a proxy server, set **enable** to **true**.
proxy_addr	Enter the IP address of the proxy server, not that of the target server. The proxy is used by the source server to access SMS. Replace *your-proxy-addr.com* with the IP address of your own proxy server. Configure the protocol based on your proxy server. HTTPS is recommended.
proxy_port	Enter the proxy port opened on the proxy server.
proxy_user	Enter the proxy username if available, for example, root. Otherwise, leave it blank.
use_password	If the proxy server has a password, set this parameter to **true**. Otherwise, set it to **false**.

Step 6 When prompted, enter the AK/SK pair for the Huawei Cloud account that you are migrating to and the SMS domain name. You can obtain the SMS domain name on the **Agents** page of the SMS console, as shown in **Figure 2-3**.

If the EPS service has been enabled for your Huawei Cloud account, after you entered the AK/SK pair, the Agent will list all enterprise projects your account is allowed to access. You can select the enterprise project you would like to migrate the source server to. This enables you to isolate permissions, resources, and finance during the migration. For details, see **Migrating a Server into an Enterprise Project**.

Figure 2-3 Obtaining the SMS domain name



After the authentication succeeds, the SMS-Agent starts to report source server information to SMS, and the window is closed. You can go to the **Servers** page on the SMS console to check the record of the source server.

----End

Troubleshooting

- **SMS.0202 AK/SK Authentication Failed**
- **SMS.1902 Failed to Start the I/O Monitoring Module**
- **Why Wasn't My Source Server Added to the SMS Console After I Configured the Agent?**

Uninstalling the SMS-Agent

After the migration is complete, you can uninstall the SMS-Agent from the source server. For details, see **Uninstalling the SMS-Agent from Windows**.

2.2 Installing the SMS-Agent on Linux

Scenarios

Before the migration, you must install the Agent on the source server to be migrated. During the installation, you need to enter the AK/SK pair of the Huawei Cloud account you are migrating to. After the Agent is started, it automatically reports source server information to SMS. The information is used for migration only. To learn what source server information is collected by SMS, see **What Information Does SMS Collect About Source Servers?**

NOTE

Before using SMS to migrate servers, you need to manually install and register the SMS-Agent on each server to be migrated. If there are more than 50 servers to migrate, you can **create a batch server migration workflow** on MgC to automate batch installation and registration of the Agent.

Constraints

You need to log in to the source Linux server as user root.

Prerequisites

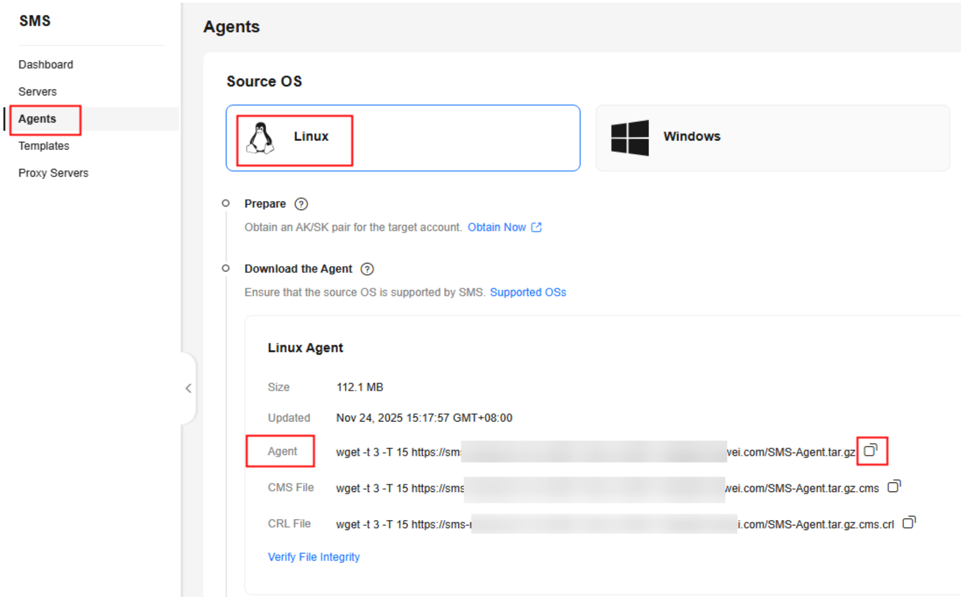
- You have obtained an AK/SK pair for your Huawei Cloud account.
  - If you use an IAM user for migration, obtain an AK/SK pair by referring to **How Do I Obtain an AK/SK Pair for an IAM User?**
  - If you use an account for migration, obtain an AK/SK pair by referring to **How Do I Obtain an AK/SK Pair for a Huawei Cloud Account?**
- You have confirmed that the source server OS is supported by SMS. Learn more about **supported Linux OSs**.

Procedure

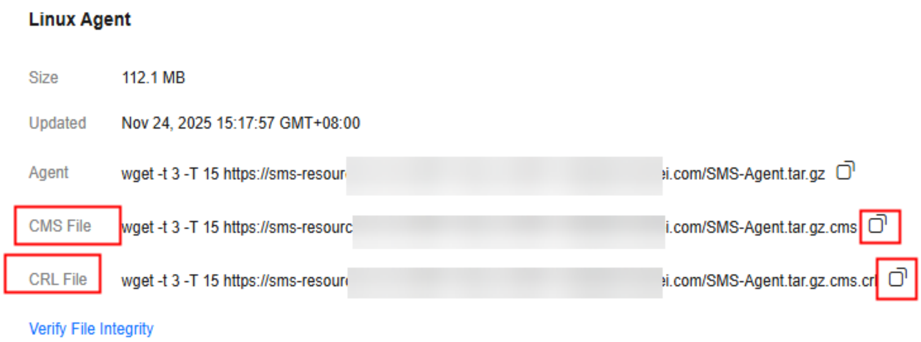
Step 1 Sign in to the **SMS console**.

Step 2 In the navigation pane, choose **Agents**. The **Agents** page is displayed.

Step 3 Select the **Linux** card, and in the **Linux Agent** area, click the icon next to Agent URL to copy the Agent download command. Run the command on the source server to download the Agent installation package.



Step 4 Copy and run the commands to download the CMS and CRL files to the Linux source server, then verify the integrity of the Agent installation file. For details, see **How Do I Verify the Integrity of the Agent Installation File?**



Step 5 Decompress the SMS-Agent software package.

```
tar -zxvf SMS-Agent.tar.gz
```

Step 6 Switch to the **SMS-Agent** directory on the source server.

```
cd SMS-Agent
```

Step 7 Determine whether to use an HTTP/HTTPS proxy server for the migration. If your source server cannot access Huawei Cloud over the Internet, you can use a proxy server. You will need to configure the proxy server yourself. In a migration over a private line or VPN, a proxy server is only used for registering the source server with SMS. It is not used for data migration.

- If you do not need an HTTP/HTTPS proxy server, skip this step and go to **the next step**.
- If you need to use an HTTP/HTTPS proxy, do as follows:
  a. Run the following command to go to the **config** directory:
  ```
  cd SMS-Agent/agent/config
  ```
  b. Open and edit the **auth.cfg** file. Do not edit the **auth.cfg** file unless you need to use an HTTP/HTTPS proxy.


```
vi auth.cfg
```

The values shown here are for reference only.

```
[proxy-config]
enable = true
proxy_addr = https://<your-proxy-address>.com
proxy_port = 3128
proxy_user = root
use_password = true
```

Configure the parameters according to the table below.

Table 2-4 Proxy parameter settings

Parameter	Description
enable	To use a proxy server, set **enable** to **true**.
proxy_addr	Enter the IP address of the proxy server, not that of the target server. The proxy is used by the source server to access SMS. Replace *your-proxy-addr.com* with the IP address of your own proxy server. Configure the protocol based on your proxy server. HTTPS is recommended.
proxy_port	Enter the proxy port opened on the proxy server.
proxy_user	Enter the proxy username if available, for example, root. Otherwise, leave it blank.
use_password	If the proxy server has a password, set this parameter to **true**. Otherwise, set it to **false**.

c. Run the following command to save the **auth.cfg** file and exit:

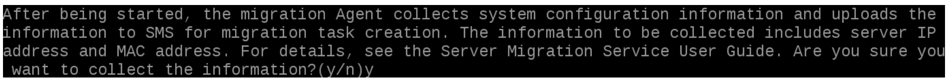
```
:wq
```

Step 8 Start the SMS-Agent.

```
./startup.sh
```

Step 9 Read the displayed information carefully, enter **y**, and press **Enter**.

Figure 2-4 Entering y



Step 10 Enter the AK/SK pair for the Huawei Cloud account that you are migrating to and the SMS domain name. You can obtain the SMS domain name on the **Agents** page of the SMS console, as shown in **Figure 2-6**.

Figure 2-5 Entering the AK/SK pair

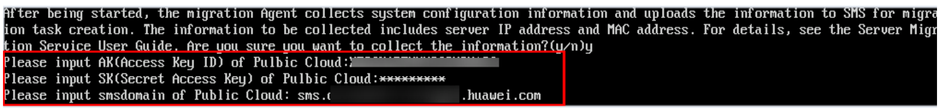
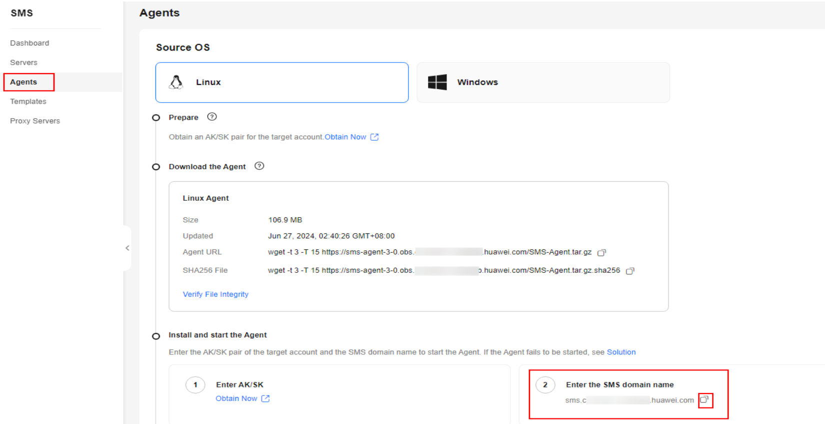


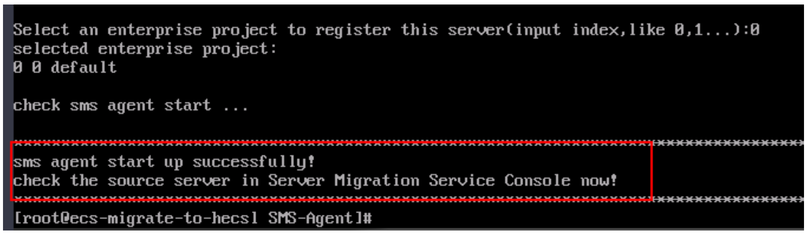
Figure 2-6 Obtaining the SMS domain name



If the EPS service has been enabled for your Huawei Cloud account, after you entered the AK/SK pair, the Agent will list all enterprise projects your account is allowed to access. You can select the enterprise project you would like to migrate the source server to. This enables you to isolate permissions, resources, and finance during the migration. For details, see **Migrating a Server into an Enterprise Project**.

When the information shown in the figure below is displayed, the SMS-Agent has been started up and will automatically start reporting source server information to SMS. You can go to the **Servers** page on the SMS console to check the record of the source server.

Figure 2-7 Agent running



----End

Troubleshooting

- **SMS.6517 rsync Not Installed on the Source Server**
- **SMS.0202 AK/SK Authentication Failed. Ensure that the AK and SK Are Correct**
- **Why Wasn't My Source Server Added to the SMS Console After I Configured the Agent?**

Uninstalling the SMS-Agent

After the migration is complete, you can uninstall the SMS-Agent from the source server. For details, see **Uninstalling the SMS-Agent from Linux**.

3 Migration Management

3.1 Overview

SMS provides multiple features to help you manage migration tasks.

Table 3-1 Features for migration task management

Feature	Description	Notes
Configuring a Target Server	Before the migration, you need to configure a target server to receive the data migrated from the source server, including the system, application, and service data. When you configure the target server, you need to specify the network, migration method, target server settings, disk and partition settings, disk encryption, and other migration settings.	You can configure the target server only when the following conditions are met: • The source server is **Connected** to SMS. • The migration is in the **Migration Feasibility Check** stage. • The migration is in the **Pending target configuration** status.
Starting a Full Replication	A full replication copies all data from the source server to the target server with a single click.	• After the full replication starts, do not restart the source server or the SMS-Agent, or the migration will fail. • During the full replication, the target server stays locked to prevent any operations. After the migration is complete, it will be automatically unlocked.

Feature	Description	Notes
Synchronizing Incremental Data	After the full replication is complete, you can manually trigger incremental synchronization to transfer newly added or modified data from the source server to the target server.	You can synchronize the incremental data from a source server only when the migration status is **Finished**.
Setting a Migration Rate	To avoid impacting source server workloads during peak business hours, you can configure time-based bandwidth limits to restrict migration traffic.	A migration rate limit must be an integer from 0 to 1,000. The unit is Mbit/s. ● If you set the rate limit to 0 or leave it empty, the migration speed is unrestricted and will follow the actual network speed between the source and target servers. ● If you set the rate limit to V1, and the actual network speed between the source and target servers is V2, the migration rate will be limited to the lower of V1 and V2.
Uploading Migration Logs	If an exception occurs, uploading run logs to an OBS bucket can provide technical support with the context needed to help you quickly identify and resolve the issue.	● Only the Agent 25.1.0 or later can upload migration logs. To view the version of the Agent you installed on a source server, you can click the source server name in the server list on the SMS console. On the displayed page, click the **Task Info** tab. In the **Source Info** area, view the Agent version. ● If a proxy server is used for the migration, you need to select an OBS bucket in the same region as the proxy server, or the logs cannot be uploaded.
Deleting Resources	After the migration is complete, if incremental data synchronization is no longer needed, you can delete the cutover and synchronization snapshots generated during the process to stop further billing.	Once these snapshots are deleted, incremental data synchronization will no longer be possible. Confirm that no additional synchronizations are required before performing this action.

Feature	Description	Notes
Deleting a Migration Task	After confirming that the migration is complete and the target server can run properly, you can delete the migration task and its configurations to release resources. Before doing so, you are advised to delete migration-related resources to avoid unnecessary expenses.	• To migrate the source server again after you delete the migration task, you can restart the SMS-Agent on the server to re-register it. • Deleting a server (or migration task) record will not delete the associated source or target server.

3.2 Configuring a Target Server

Before the migration, you need to configure a target server. The target server is used to receive the data migrated from the source server, including the system, application, and service data.

When you configure the target server, you need to specify the network, migration method, target server settings, disk and partition settings, disk encryption, and other migration settings.

Prerequisites

- You have prepared the account, required permissions, source and target servers, and network environment. For details, see **Preparing for Migration**.
- You have installed and started the Agent on the source server, and the source server is displayed in the server list on the SMS console. The source server meets the following requirements:
  - It is **Connected** to SMS.
  - The migration is in the **Migration Feasibility Check** stage.
  - The migration is in the **Pending target configuration** status.

Constraints

Before setting the target server, read the constraints and limitations of SMS. For details, see **Server Requirements** and **Migration Constraints**.

Procedure

Item	Description
Configuring Basic Settings	Select the region, port, and network type of the target server.

Item	Description
Configuring Specifications	Configure the specifications of the target server based on the specifications of the source server. There are two methods for configuring target servers. ● **Use existing**: Select a Huawei Cloud ECS as the target server. CAUTION To ensure that the target server can start properly after the migration is complete, its disks will be formatted, and its registry and network settings will be updated. You are advised to back up data before the migration. ● **Create new**: SMS automatically purchases a target server based on the configured specifications.
(Optional) Setting the Resource Limits	Set network parameters, such as the traffic limit and overspeed threshold. Note that some parameters are supported only on Linux servers.
Starting a Migration Drill	Migration drills help you fully assess the feasibility and identify potential risks of a migration task beforehand.
Configuring Migration Parameters	Set operations after migration, for example, choose whether to enable incremental data synchronization after full replication and whether to verify data consistency.
Saving the Settings and Starting the Migration Task	Confirm the parameter settings, save the settings, and start the migration.

Configuring Basic Settings

Step 1 Sign in to the **SMS console**.

Step 2 In the navigation pane, click **Servers**.

Step 3 In the server list, locate the source server to be migrated, and click **Configure Target** in the **Migration Stage/Status** column or choose **More** > **Configure Target** in the **Operation** column.

NOTE

If the source server record is not found in the server migration list, see **Why Wasn't My Source Server Added to the SMS Console After I Configured the Agent?**

Step 4 On the **Configure Basic Settings** page, configure parameters by referring to **Table 3-2**.

Table 3-2 Parameter description

Parameter	Sub-Parameter	Description
Target Region	-	Select the region for the target server.

Parameter	Sub-Parameter	Description
Target Port	-	● By default, ports 22, 8900, and 8899 are enabled for Windows. ● For a Linux server, port 22 is enabled by default for file-level migration, and ports 22 and 8900 are enabled by default for block-level migration. **NOTE** Open ports are used for the following purposes: ● Port 22: The initialization port of the transmission link. It is used to establish the transmission channel and can be modified in some regions. ● Port 8899: The control port for data transmission. It is used to transmit task control signals. It cannot be modified. ● Port 8900: The port for block data transmission. It cannot be modified. Pay attention to the following security notes for migration ports: During a migration, SMS uses the required ports to establish a transmission channel between the source server and the target server. To reduce security risks during the migration, SMS include the following built-in security settings: ● Agent image: used only during the migration and automatically uninstalled after the migration. ● SSL login: Password-based SSH login is disabled for the target server.Only certificate- or key-based SSH login is allowed. ● SSH port: The default is port 22, which can be changed in some regions. In the security group of the target ECS, allow only the required migration ports (22, 8899, and 8900), and allow access only from the IP address of the source server. Do not enable other unnecessary ports or set the source IP address to 0.0.0.0/0.

Parameter	Sub-Parameter	Description
		CAUTION ● Strictly comply with the requirements above; otherwise the target ECS may be accessed by unauthorized IP addresses over unnecessary ports, increasing the risk of being scanned, cracked, or attacked. You are responsible for any security risks caused by incorrect security group rules. ● Use the latest stable version of rsync if needed, and complete security configurations according to the official rsync security hardening guide. You are responsible for any security risks resulting from using an earlier version, a version with known vulnerabilities, or a version not configured in accordance with official security requirements.
Network Type	Public	(Default) Migration over the Internet requires that the target server has an EIP bound.
	Private	A Direct Connect connection, VPN connection, VPC peering connection, VPC subnet, or Cloud Connect connection must be provisioned. The private IP address of the target server will be used for migration. For details about the network scenarios and solutions for migration over a private network, see **Setting Up an SMS Migration Network Using VPN, Direct Connect, and Cloud Connect**.
IP Version	IPv4	IPv4 is used for data migration by default.
	IPv6	On a dual-stack network, IPv6 can be used for migration. For details about migration over IPv6, see **Migrating Servers over an IPv6 Network**. **CAUTION** If IPv6 is selected, you can only select an existing server as the target end, instead of creating one.

----End

Configuring Specifications

There are two methods for configuring target servers.

- **Use existing**: Select a server created on Huawei Cloud as the target server.

⚠ CAUTION

To ensure that the target server can start properly after the migration is complete, its disks will be formatted, and its registry and network settings will be updated. You are advised to back up data before the migration.

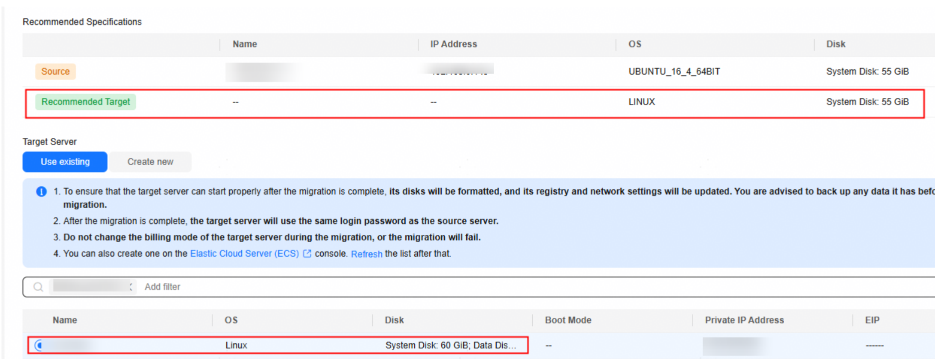
- **Create new**: SMS automatically purchases a target server based on the configured specifications.

For details about the requirements on target servers, see **Target Server Requirements**.

Selecting an Existing Server

In the list of existing servers, select one that meets the specifications requirements displayed in the **Recommended Target** row. If no existing server meets the requirements, click **Elastic Cloud Server (ECS)** above the list and purchase an ECS based on the recommended specifications. For details, see **Purchasing an ECS in Custom Config Mode**. You can select a pay-per-use or yearly/monthly ECS.

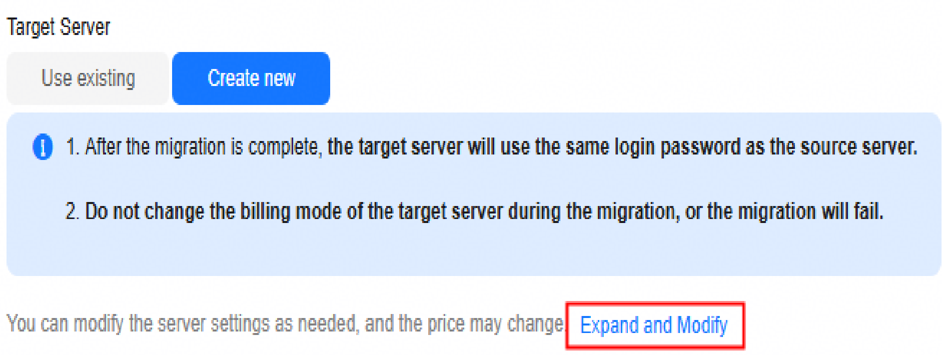
Figure 3-1 Select an existing target server



Creating a Server

If you select **Create new**, the system automatically recommends server specifications, including the server name, AZ, VM specifications, disk information, EIP, VPC, subnet, and security group. You can modify these settings as needed.

Figure 3-2 Modifying server configurations



- If you select **Recommend** for **Server Template**, the system automatically creates the AZ, instance specifications, disk type, VPC, subnet, and security group during the migration. You can adjust the settings.

  NOTE

  - If **Create new** is selected for **VPC**, the system automatically creates a VPC for the target server based on the following rules:
    - If the source server's IP address is 192.168.*X*.*X*, the system creates a VPC and a subnet that both belong to network range 192.168.0.0/16.
    - If the source server's IP address is 172.16.*X*.*X*, the system creates a VPC and a subnet that both belong to network range 172.16.0.0/12.
    - If the source server's IP address is 10.*X*.*X*.*X*, the system creates a VPC and a subnet that both belong to network range 10.0.0.0/8.
  - If **Create new** is selected for **Security Group**, the system automatically creates a security group for the target server and allows traffic to the target server over certain ports:
    - Windows: ports 8899, 8900, and 22
    - Linux: port 22 for file-level migration
    - Linux: ports 8900 and 22 for block-level migration
- If you select **Custom Template** for **Server Template**, the system automatically sets the AZ, instance specifications, disk type, VPC, subnet, and security group based on the template. You can adjust the settings. For details about how to create a custom template, see **Creating a Server Template**.
- Configure advanced disk settings.
  - Data disks must be either VBD or SCSI. VBD is the default device type for data disks. For details about disk device types, see **Device Types**.
  - Data disks can be created as shared disks. For details about shared disks, see **Managing Shared EVS Disks**.
  - For target servers newly created by the system, system and data disks can be encrypted. For details about shared disks, see **Managing Shared EVS Disks**. To enable disk encryption, you need to create an agency to authorize EVS to access KMS. After the authorization is successful, set **KMS Encryption** to one of the following values:
    - **Select an existing key**

      Select a key from the drop-down list. You can select one of the following keys:

      Default keys: After the KMS access permissions have been granted to EVS, the system automatically creates a default key and names it **evs/default**.

      Custom keys: You can choose an existing key or create one. For details about how to create a key, see **Creating a Key**.
    - **Enter a key ID**

      Enter the ID of a key shared from another user. Ensure that the key is in the target region. For details, see **Creating a Grant**.

NOTE

- Before the migration is complete, do not disable or delete the key used, or the migration will fail.
- The encryption attribute of a disk cannot be modified after the disk is created.
- Keys can be shared with accounts, not IAM users.
- If KMS encryption is used, you will be billed for what you use beyond the free quota given by KMS. For details, see **DEW Billing Overview**.

(Optional) Setting the Resource Limits

Set the parameters according to **Table 3-3**.

Table 3-3 Resource limit parameters

<table>
<tr><th>Parameter</th><th>Description</th></tr>
<tr><td>CPU Limit</td><td rowspan="3">These options are only available for Linux migrations. For details, see How Do I Limit Resource Allocation for the Agent in a Linux Migration?</td></tr>
<tr><td>Memory Limit</td></tr>
<tr><td>Disk Throughput Limit</td></tr>
<tr><td>Migration Rate Limit</td><td>You can limit the migration rate based on the source bandwidth and service requirements. If you do not want to limit the migration rate, set this parameter to 0.
Traffic limiting is unavailable if:
• The migration uses an IPv6 network.
• Traffic Control (TC) is missing from the source server.
• The TC module is available, but the Class-Based Queuing (CBQ) or Hierarchy Token Bucket (HTB) module is missing.</td></tr>
<tr><td>Overrate Threshold (%)</td><td>You can regulate how much the migration rate can exceed the configured limit. If the migration rate exceeds the threshold multiple consecutive times, the migration task is automatically paused.
For example, if the migration rate limit is set to 10 Mbit/s and the overrate threshold is set to 10%, the task is automatically paused when the migration rate exceeds 11 Mbit/s (110% of the limit) multiple times consecutively.
CAUTION
This option is only available for Linux migrations. It will not be available or applied if:
• The migration uses an IPv6 network.
• Traffic Control (TC) is missing from the source server.
• The TC module is available, but the Class-Based Queuing (CBQ) or Hierarchy Token Bucket (HTB) module is missing.
• The installed SMS-Agent is earlier than 24.9.0.</td></tr>
</table>

Starting a Migration Drill

Migration drills help you fully assess the feasibility and identify potential risks of a migration task beforehand. The system verifies if security group ports are set correctly, domains connect normally, and necessary permissions are available. If issues are identified, the system offers remediations to minimize risks and disruptions during migration.

After this function is enabled, the system automatically launches a migration drill before starting the full replication. The entire migration drill takes 5 to 15 minutes, and pay-per-use resources incur costs during this period. For details, see **Billing**. A migration drill does not transmit source data, so the fee incurred in the drill is low.

You can review the drill results in the task details. For details, see **Checking the Migration Drill Status and Report**.

Configuring Migration Parameters

Set the parameters according to **Table 3-4**.

NOTE

The parameters for Windows and Linux servers are different. Set their parameters accordingly.

Table 3-4 Migration settings

Parameter	Sub-Parameter	Description
Start Target Upon Launch	No	The target server will be stopped after the migration is complete.
	Yes	The target server will be started after the migration is complete.
Migration Method	Linux block-level	Migration and synchronization are performed by block. This method is efficient, but the compatibility is poor.
	Linux File-Level	Migration and synchronization are performed by file. This method is inefficient, but the compatibility is excellent. File-level migration is used by default for Linux OSs.
	Windows Block-Level	Migration and synchronization are performed by block. Block-level migration is used by default and cannot be changed.
Enable Concurrency	Automatic	The SMS-Agent automatically configures the maximum number of migration processes allowed based on source server conditions.

Parameter	Sub-Parameter	Description
	Manual	You can specify the maximum number of processes that the SMS-Agent can start concurrently for migration and synchronization tasks, respectively. This option is only available for Linux file-level migrations. For more information, see **How Do I Set the Number of Concurrent Processes for a Linux File-Level Migration?**
Enable Continuous Synchronization	-	Disabled: After full replication is complete, the system automatically starts the target server without synchronizing incremental data from the source server. You need to choose **More** > **Synchronize** in the **Operation** column to synchronize incremental data to the target server.
		Enabled: After full replication is complete, the system automatically starts to synchronize incremental data from the source end to the target end periodically. In this case, the target server is not started and cannot be operated. To exit this phase, click **Start Target** in the **Migration Stage/ Status** column of the migration task to start the target server.
Verify Data Consistency	-	Disabled: Data consistency is not verified after full replication is complete. You can choose whether to perform data consistency check during incremental synchronization.

<table>
<tr><th>Parameter</th><th>Sub-Parameter</th><th>Description</th></tr>
<tr><td></td><td></td><td>Enabled: Data consistency is automatically verified after full replication is performed. This is a quick verification, and only the file size and last modification time will be verified. You can modify the verification policy when you launch an incremental synchronization. Note that consistency verification cannot be performed for servers with Btrfs file systems. The following describes the parameters for data consistency verification:
● Enable Hash Verification: If this option is enabled, the system will generate and compare hash values for each file to be verified. Hash verification is recommended when individual files are large and important. Enabling this option will increase CPU and disk I/O overheads for the source server and extend the verification time.
CAUTION
● Hash values cannot be calculated for files in use, so these files will be skipped during the verification.
● Enabling this option requires you to specify the verification scope, and only files in the specified scope will be verified.
● Verification Scope
– Under Exclude paths, enter the paths you want to exclude from the verification. A maximum of 30 paths can be entered. Use commas (,) to separate the paths. For example, /root/data,/var. Leaving it empty will initiate a full verification.
– Under Include paths, enter the paths you want to verify.</td></tr>
</table>

Parameter	Sub-Parameter	Description

		CAUTION
		● If the entered paths are incorrect or empty, 0 will be displayed for them in the verification results.
		● The more data you need to verify, the longer the verification will take. It is wise to narrow the verification scope to only key paths.
		● The following paths will be excluded from consistency verification by default:
		  ● Linux: **/bin**, **/boot**, **/dev**, **/home**, **/etc**, **/lib**, **/media**, **/proc**, **/sbin**, **/selinux**, **/sys**, **/usr**, **/var**, **/run**, and **/tmp**
		  ● Windows: top-level directories of partitions, for example, **C:** and **O:**
		If you need to include any of the preceding excluded paths in the verification, refer to **Modifying the Default Excluded Paths**.

Resize Disks and Partitions	-	Disabled: The number of disks and partition size of the target server are the same as those of the source server.
		Enabled: You can adjust the number of target disks and partition size. For details, see **Resizing Disks and Partitions**.
Transit IP Address	-	This parameter is available only for private line migration. It is used to set the transit IP address of the target server. For details about how to configure the network in this scenario, see **For a Scenario Where the Source Server Has No Internet Access and Cannot Communicate with the Target Server**.

Saving the Settings and Starting the Migration Task

After configuring the target server settings, review them in the summary panel and choose whether to start the migration now or later.

- Saving the settings and starting the task immediately
  a. Click **Save and Start**.
  b. In the displayed dialog box, read the migration conditions and click **OK**. Return to the migration server list. The task status becomes **Running**, indicating that the migration has started.
- Saving the settings and starting the task later
  a. After you confirm that the settings are correct, click **Save**.
  b. In the displayed dialog box, read the migration conditions and click **OK**. Return to the migration server list. The task status becomes **To be started**, indicating that the migration is not started.

To start the migration, click **Start** in the **Migration Stage/Status** column.

Resizing Disks and Partitions

You can choose whether to migrate specific source partitions and then resize the paired target partitions as needed.

Resizing Disks and Partitions on Windows

The restrictions on resizing Windows disks and partitions are as follows:

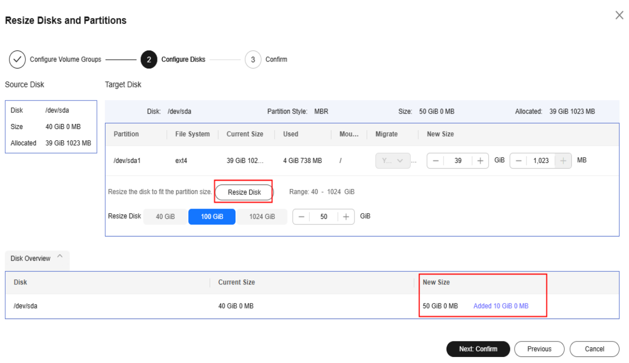
- The system and boot partitions on Windows cannot be resized.
- For a Windows server, you can upsize partitions, but you cannot downsize them.

Step 1 In the **Migration Settings** area, enable **Resize Disks and Partitions** and click **Resize Disks and Partitions**.

Step 2 Click **Resize Disk** and adjust the size of each disk as required. As shown in **Resizing disks and partitions on Windows**, you can view the resized disks at the bottom of the window.

- If the total partition size after resizing is larger than the disk size, you need to expand the disk capacity to fit the partition size.
- If the total partition size after resizing is much smaller than the disk size, you can downsize the disk.

Figure 3-3 Resizing disks and partitions on Windows



Step 3 Click **Next: Confirm**. Confirm the configurations and click **OK**.

CAUTION

Once disks and partitions are resized, this function cannot be undone directly. To restore your original settings, locate the resizing task and choose **More** > **Delete** in the **Operation** column, and then restart the Agent on the source server to reconfigure your target server parameters.

----End

Resizing Disks and Partitions on Linux

The restrictions on resizing Linux disks and partitions are as follows:

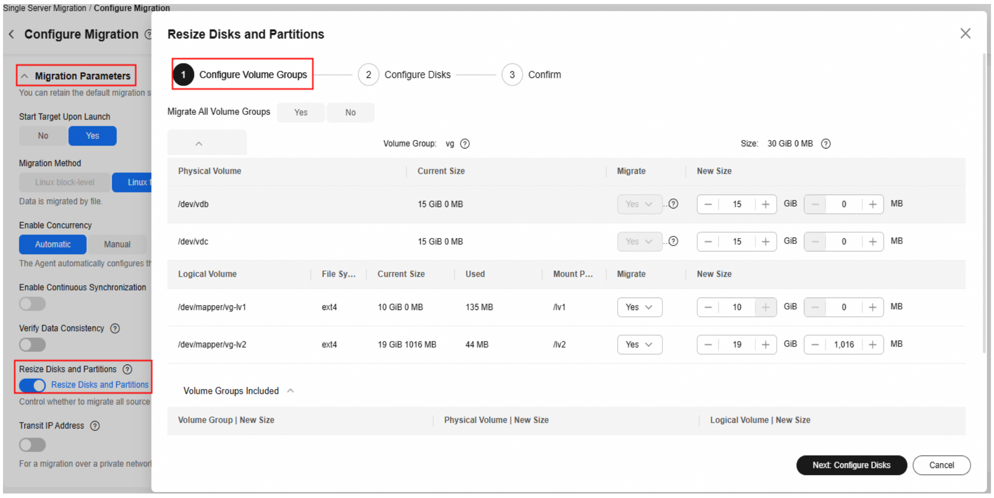
- For a Linux server using LVM, you can choose whether to migrate specific physical or logical volumes and resize the paired target volumes.
- Partition resizing is not available for Btrfs partitions on Linux.
- On a Linux system, the swap partition is migrated by default and cannot be changed.
- You can choose to migrate all or none volume groups by using the **Migrate All Volume Groups** option.
- If you choose to migrate none of the logical volumes in a volume group, their physical volumes will not be migrated by default.
- In a Linux block-level migration, you can upsize partitions, but you cannot downsize them.
- In a Linux file-level migration, you can upsize or downsize partitions. When you downsize a partition, the new partition size must be at least 1 GB larger than the used partition space. If the current size does not meet this condition, downsizing is not possible. For details, see **What Are the Rules for Resizing Volume Groups, Disks, and Partitions?**

Step 1 In the **Migration Settings** area, enable **Resize Disks and Partitions** and click **Resize Disks and Partitions**.

Step 2 Configure volume groups.

- If the source server uses LVM, you can resize logical and physical volumes.
- If the source server does not use LVM, skip this step.

Figure 3-4 Configuring volume groups

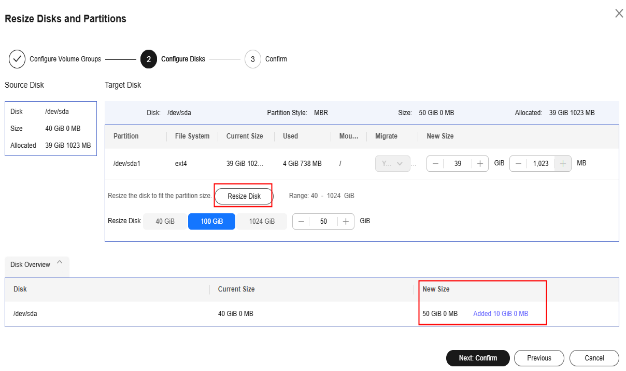


Step 3 Click **Next: Configure Disks**.

Step 4 Click **Resize Disk** and adjust the size of each disk as required. As shown in **Resizing disks and partitions on Linux**, you can view the resized disks at the bottom of the window.

- If the total partition size after resizing is larger than the disk size, you need to expand the disk capacity to fit the partition size.
- If the total partition size after resizing is much smaller than the disk size, you can downsize the disk.

Figure 3-5 Resizing disks and partitions on Linux



Step 5 Click **Next: Confirm**. Confirm the configurations and click **OK**.

> **CAUTION**
>
> Once disks and partitions are resized, this function cannot be undone directly. To restore your original settings, locate the resizing task and choose **More** > **Delete** in the **Operation** column, and then restart the Agent on the source server to reconfigure your target server parameters.

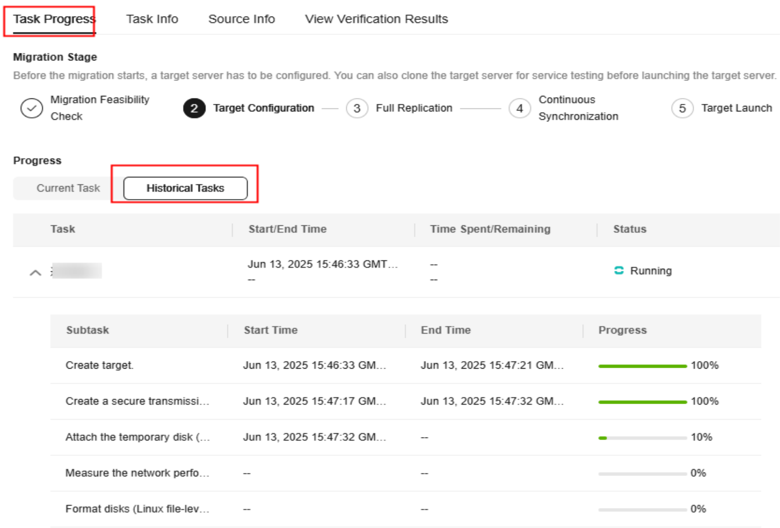
----End

Checking the Migration Drill Status and Report

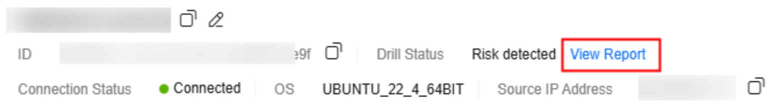
If **Migration Drill** is enabled when you configure a migration task, you can check the migration drill status and report.

Step 1 In the server list, click the name of the source server to expand the migration task details.

Step 2 On the **Task Progress** tab, check the status and progress of the migration drill.



Step 3 In the upper part of the task details page, click **View Report** next to **Drill Status**. In the report displayed on the right, check the drill details, check items, and check results.



----End

Helpful Links

If an error occurs during the configuration of the target server, see **Target Server Configuration** for troubleshooting.

3.3 Starting a Full Replication

A full replication replicates all data from the source server to the target server. The replication speed depends on the outbound bandwidth of the source server or the inbound bandwidth of the target server, whichever is smaller.

Constraints

- Before a migration is complete, **do not** perform operations on the OS or disks of the target server, including but not limited to **changing or reinstalling the OS**.
- **Do not** change the billing mode of the target server or its disks to Yearly/ Monthly during a migration.
- Make sure you have backed up any data on the target server that you need to save and ensure that the disks can be formatted. Disks on the target server will be formatted and re-partitioned based on the source disk settings during the migration.
- During the migration of a Windows source server, **do not restart the source server**. The restart will disconnect the source server from the SMS console. To

retry the migration, you need to delete the migration task and create a new one.

- Do not restart the SMS-Agent during a Windows migration or a Linux block-level migration.
- You are advised not to start the SMS-Agent during a Linux file-level migration unless necessary, even though resumable data transfer is supported.
- During the migration, a temporary pay-per-use disk is created and attached to the target server to ensure that the migration runs normally. Once the migration is complete, the disk is automatically detached and deleted. If you manually delete the migration task before it completes, you need to manually delete the temporary disk to avoid extra expenses.
- If an error occurs during the migration, you are advised to provide migration logs to technical support engineers so that the fault can be resolved quickly.

  For details about how to get migration logs, see **Where Can I Find the Agent Run Logs?**

Prerequisites

- The target server has been configured. For details, see **Configuring a Target Server**.
- The source server status meets the following requirements: **Migration Stage** is **Full Replication**, and **Migration Status** is **Ready**.

Starting Full Replication

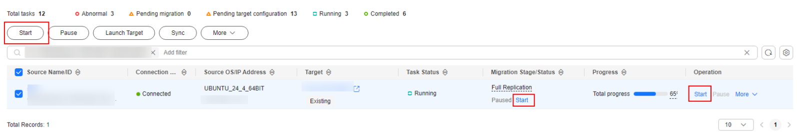
Step 1 Sign in to the **SMS console**.

Step 2 In the navigation pane, click **Servers**.

Step 3 In the server list, locate the server to be migrated and start the migration in either of the following ways:

- Click **Start** in the **Migration Stage/Status** or **Operation** column. In the displayed **Start Migration** dialog box, click **OK** to start full replication.
- Select the source server and click **Start** above the server list. In the displayed **Start Migration** dialog box, click **OK** to start full replication.

Figure 3-6 Starting a full replication



NOTE

During the full replication, the target server stays locked to prevent any operations. After the migration is complete, it will be automatically unlocked.

Step 4 In the server list, click the name of the source server to view the migration progress.

Step 5 Wait for the full replication to complete.

- If you set **Enable Continuous Synchronization** to **No** when you defined the migration settings, after the full replication is complete, the system puts the migration to a **Target Launch** stage and launches the target server to complete the migration automatically.
- If you set **Enable Continuous Synchronization** to **Yes** when you defined the migration settings, after the full replication is complete, the migration will move on to a **Continuous sync** status, and any new or modified data will be automatically synchronized from the source server to the target server. You will need to manually launch the target server to complete the migration. For details, see **Launching a Target Server**.

----End

Related Operations

- After the migration and service cutover are complete, you need to adjust the configurations of the target server based on service requirements. For details, see **What Configuration Items Need to Be Manually Modified After a Server Is Migrated?**
- After the migration is complete, if you confirm that no additional incremental synchronizations are required, you can delete the snapshots generated for cutover and synchronization during the migration. For details, see **Deleting Resources**. If you do not delete the resources, the system will retain the snapshots generated for cutover and synchronization for an extended period of time. The retained snapshots will continue to incur charges in certain regions.

3.4 Synchronizing Incremental Data

After the target server is launched, if there are data changes on your source server, you can synchronize the incremental data from the source server to the target server.

The data changes on the target server will be overwritten by the data synchronized from the source server. For details, see **Will an Incremental Synchronization Overwrite Existing Data on a Launched Target Server?**

Constraints

- You can synchronize the incremental data from a source server only when the migration status is **Finished**.
- During a Linux file-level migration, if the source server has a large number of files in a single partition and has limited CPU and memory resources (for example, 1 vCPU and 1 GB memory), rsync may fail to perform incremental synchronization due to insufficient resources. In this case, you are advised to perform only the initial full migration and avoid performing incremental synchronization.
- After the full migration of a Linux server, some parameter settings in the directories below on the target server are modified to make the target server compatible with Huawei Cloud and ensure that the target server can start properly. During incremental synchronization, data in these directories is not synchronized by default to prevent parameter settings in these directories on the target server from being overwritten or modified.


```
/proc/*,
/sys/*,
/lost+found/*,
/tmp/_MEI*,
/var/lib/ntp/proc/*,
/
/etc/fstab,
/etc/X11/*,
/root/initrd_bak/*,
/lib/modules/*,
/boot/grub2/x86_64-efi/*,
/boot/grub2/i386-pc/*
```

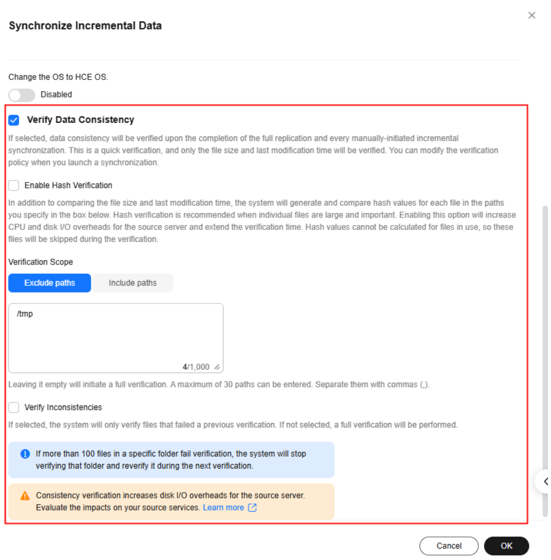
Synchronizing Incremental Data

Step 1 Sign in to the **SMS console**.

Step 2 In the navigation pane, click **Servers**.

Step 3 In the server list, locate the source server you want to synchronize, and click **Sync** above the list, or choose **More** > **Sync** in the **Operation** column.

Step 4 In the **Synchronize Incremental Data**dialog box, carefully read the tips, enable **Verify Data Consistency** if needed, and click **OK**. For details about this option, see **How Do I Verify Data Consistency Between the Source and Target Servers?**



Step 5 Then click **OK**. Wait until the task status changes to **Completed**, which indicates that the incremental data has been synchronized.

If data synchronization is no longer required, you can delete the cutover snapshots and synchronization snapshots generated during the migration. For details, see **Deleting Resources**.

----End

Troubleshooting

Common synchronization issues and their solutions:

- **SMS.0806 Failed to Synchronize Partition to Target Server**

- **SMS.1414 Migration Module Stopped Abnormally and Cannot Synchronize Data**

3.5 Setting a Migration Rate

A large amount of traffic and bandwidth will be consumed during migration. To avoid impacting source server workloads during peak business hours, you can configure time-based bandwidth limits to restrict migration traffic.

Constraints

- The source server is **Connected** to SMS.
- The migration rate limit must be an integer from 0 to 1,000.
- If you set the rate limit to 0 or leave it empty, the migration speed is unrestricted and will follow the actual network speed between the source and target servers.
- The migration rate is bottlenecked by the migration rate limit you configure or the actual network speed, whichever is smaller.

Requirements for Migration Speed Limiting on Linux

To limit the migration speed for a Linux source server, the **tc** command and **cbq** or **htb** kernel module must be installed on the source server.

1. To check whether the **tc** command is installed, run the following command on the Linux terminal. If the system returns the tc function list, the **tc** command has been installed.
```
# tc
```
2. To check whether the **cbq** module is installed, run the following command on the Linux terminal. If the system returns the related information, the **cbq** module is loaded.
```
lsmod | grep sch_cbq
```
3. If the **cbq** module does not exist, check the **htb** module. Run the following command on the Linux terminal. If the system returns the related information, the **htb** module is loaded.
```
lsmod | grep sch_htb
```

Setting a Migration Rate

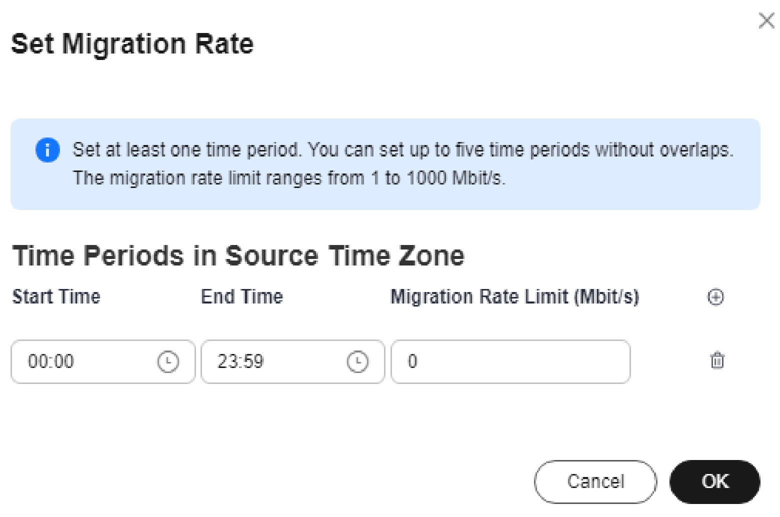
Step 1 Sign in to the **SMS console**.

Step 2 In the navigation pane, click **Servers**.

Step 3 Locate the server for which you want to set the migration rate, and choose **More** > **Set Migration Rate** in the **Operation** column.

Step 4 In the **Set Migration Rate** dialog box, set a period and rate limit.

Figure 3-7 Setting migration rate limits



Step 5 Click **OK**. If **Setting migration rate...** is displayed, the setting is complete. Check whether the rate limit takes effect in the specified period.

----End

Helpful Links

For FAQs about the migration rate and duration, see:

- **How Is the Migration Rate Displayed on the SMS Console Calculated?**
- **What Factors Affect the Migration Speed?**
- **Why Is the Linux Block-Level Migration Very Slow?**

3.6 Uploading Migration Logs

SMS allows you to upload migration logs to a specified OBS bucket for quick troubleshooting. You can repeatedly upload logs after the full replication is started.

Constraints

- Only the Agent 25.1.0 or later can upload migration logs. To view the version of the Agent you installed on a source server, you can click the source server name in the server list on the SMS console. On the displayed page, click the **Task Info** tab. In the **Source Info** area, view the Agent version.
- If a proxy server is used for the migration, you need to select an OBS bucket in the same region as the proxy server, or the logs cannot be uploaded.

Important Notes

You will be billed for storing logs in OBS.

Prerequisites

You have **created an OBS bucket** before uploading migration logs.

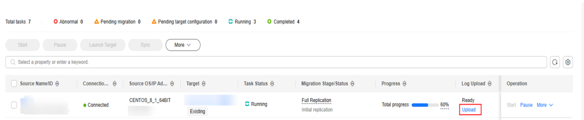
Uploading Migration Logs

Step 1 Sign in to the **SMS console**.

Step 2 In the navigation pane, click **Servers**.

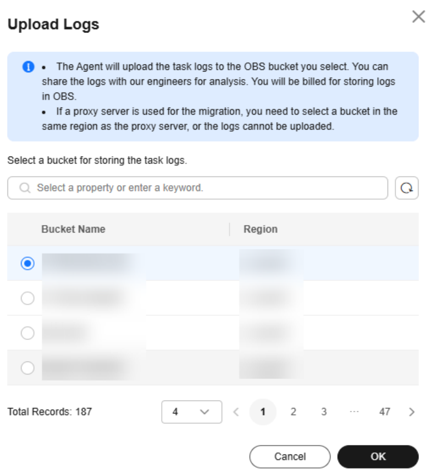
Step 3 Locate your server and click **Upload** in the **Log Upload** column.

Figure 3-8 Uploading logs



Step 4 In the displayed **Upload Logs** dialog box, select an OBS bucket for storing the uploaded logs and click **OK**. Then logs will be uploaded, and **Uploading** will appear in the **Log Upload** column.

Figure 3-9 Selecting an OBS bucket

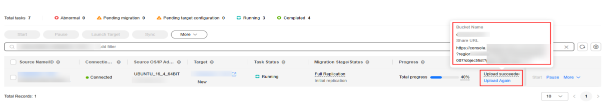


NOTE

The log upload status can be **Not ready**, **Ready**, **Uploading**, **Upload succeeded**, or **Upload failed**.

Step 5 Check whether **Upload succeeded** appears. If it does, the migration logs have been successfully uploaded. After migration logs are uploaded, move the cursor to **Upload succeeded** to check the name of the OBS bucket where the logs are stored and the sharable URL.

Figure 3-10 Logs uploaded



Step 6 (Optional) Click next to **Share URL**, and copy and paste the access URL to a browser to check the logs.

----End

3.7 Deleting Resources

After the migration is complete, if incremental data synchronization is no longer needed, you can delete the cutover and synchronization snapshots generated during the process to stop further billing.

CAUTION

Once these snapshots are deleted, incremental data synchronization will no longer be possible. Confirm that no additional synchronizations are required before performing this action.

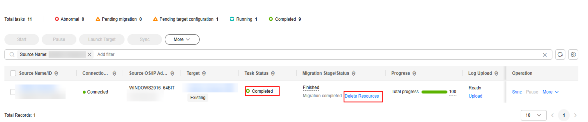
Deleting Resources

Step 1 Sign in to the **SMS console**.

Step 2 In the navigation pane, click **Servers**.

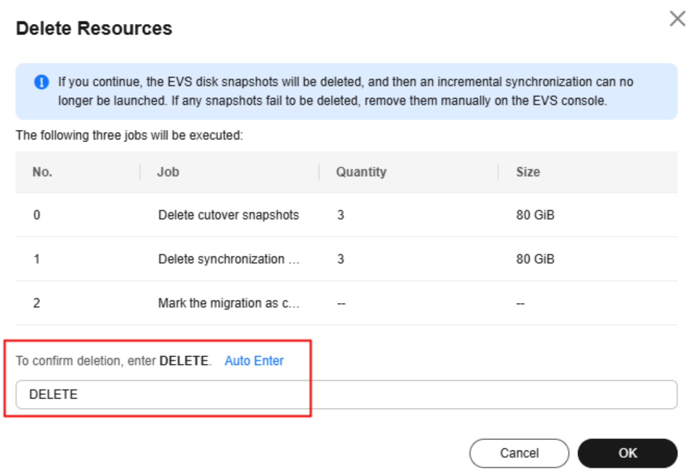
Step 3 Select a server whose task status is **Completed** and for which incremental synchronization is no longer required. Click **Delete Resources** in the **Migration Stage/Status** column.

Figure 3-11 Deleting resources



Step 4 In the displayed **Delete Resources** dialog box, enter **DELETE** in the text box or click **Auto Enter** and then click **OK**.

Figure 3-12 Confirm the deletion



Step 5 After the deletion is complete, go to the **EVS console** to check the deletion results. If the EVS snapshots fail to be deleted, manually delete them on the EVS console.

----End

Helpful Links

- **Billing for EVS Snapshots**
- **What Are the Snapshots of a Target Server Used for?**
- **View the List of Snapshots**

3.8 Deleting a Migration Task

Once the migration is complete, the target server runs properly, and incremental synchronization is no longer required, you can delete the migration task and its configurations to clean up your console.

Before deleting the task, you are advised to **delete migration-related resources** to avoid unnecessary expenses.

> ⚠ CAUTION
>
> Deleting a server migration record only deletes the record registered with SMS. It does not delete the source or target server. To register the source server again, restart the SMS-Agent on the source server.

Deleting a Task

Step 1 Sign in to the **SMS console**.

Step 2 In the navigation pane, click **Servers**.

Step 3 Locate the migration record you want to delete, and choose **More** > **Delete** in the **Operation** column.

You can also select the record and choose **More** > **Delete** in the upper left corner of the server list.

Step 4 Read the precautions carefully in the **Delete Server** dialog box. After confirming the deletion, enter **DELETE** in the text box and click **OK**.

If the server record disappears from the server list, the deletion is successful. After the deletion is complete, go to the **EVS console** to check whether the temporary disk and synchronization snapshots generated during the migration are also deleted. If the temporary resources fail to be deleted, manually delete them.

----End

4 Target Server Management

4.1 Overview

SMS provides multiple features to help you manage target servers.

Table 4-1 Features for target server management

Feature	Description	Notes
Cloning a Target Server	To verify whether your target server can run properly before launching it, you can clone it as a new ECS in the continuous synchronization stage for service pre-verification.	The cloned server must be in the same AZ as the target server but can be in a different VPC.
Launching a Target Server	To stop continuous synchronization of incremental data from the source server to the target server, you can manually launch the target server to complete the migration. Once the target server is launched, the system terminates continuous synchronization of incremental data.	To synchronize newly generated incremental data, you can click **Sync** in the **Operation** column.

Feature	Description	Notes
Viewing Server Details	To review key metrics such as the migration rate, amount of migrated data, and migration progress, you can visit the server details page. This page also displays the source and target server information, migration status, and details of any errors that occurred.	The source and target server information is collected by SMS only for data migration. To learn what source server information is collected by SMS, see **What Information Does SMS Collect About Source Servers?**
Deleting the Target Server Configuration	To update the target server or migration settings, you can delete the target server configuration from the task. Then you can re-configure the target server.	After the target server configuration is deleted from a migration task, incremental data synchronization will no longer be possible. If you reset the target server and migrate data again, all data on the source server will be migrated again, overwriting the existing data on the target server.
Deleting a Server Clone	If the ECS cloned from the target server has been tested, you can delete the ECS to release resources.	After you delete the ECS on the SMS console, you can check whether the deletion is successful on the ECS console.

4.2 (Optional) Cloning a Target Server

Before launching a target server, you can clone the target server for service testing, and only launch the target server after tests confirm there are no issues.

Constraints

The server clone must be in the same AZ as the target server, but can be in a different VPC.

Prerequisites

The migration task is in the **Continuous sync** stage.

Cloning a Target Server

Step 1 Sign in to the **SMS console**.

Step 2 In the navigation pane, click **Servers**.

Step 3 Locate the target server you want to clone, choose **More** > **Manage Target** > **Clone Target** in the **Operation** column.

Step 4 Set the parameters and click **Clone Target**.

- If you select **Recommended** for **Server Template**, the system automatically sets **VPC**, **Subnet**, **Security Group**, and parameters in **Advanced Settings** based on the current target server configuration. You can modify these parameters.
- If you select an existing template for **Server Template**, parameters **VPC**, **Subnet**, **Security Group**, and those in **Advanced Settings** are determined by the template. You can modify these parameters.

----End

4.3 Launching a Target Server

If you set **Enable Continuous Synchronization** to **Yes** when configuring the migration settings, you need to manually launch the target server after the full replication is complete. If you set **Enable Continuous Synchronization** to **No**, skip this section as the system will automatically launch the target server after the full replication is complete.

You can launch the target server when the migration is in the **Continuous sync** status. Then continuous synchronization will be interrupted. After the target server is launched, you can start an incremental synchronization by clicking **Sync** in the **Operation** column.

Before launching a target server, you can clone the target server for service testing, and only launch the target server after tests confirm there are no issues.

NOTE

The server clone must be in the same AZ as the target server, but can be in a different VPC.

Launching a Target Server

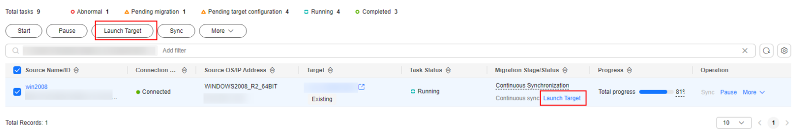
Step 1 Sign in to the **SMS console**.

Step 2 In the navigation pane, click **Servers**.

Step 3 Locate the target server you want to launch, and click **Launch Target** in the **Migration Stage/Status** column.

Alternatively, select the server that has been replicated and continuously synchronized, and click **Launch Target** above the server list.

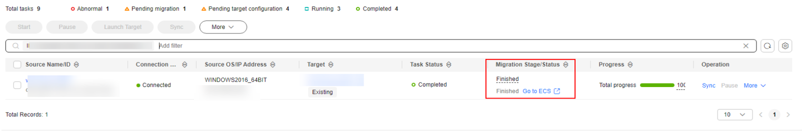
Figure 4-1 Launch Target



Step 4 In the displayed **Launch Target** window, click **OK**.

If **Finished** appears in the **Migration Stage/Status** column, the target server has been launched and the migration is complete.

Figure 4-2 Completed migration



Step 5 Click the server name in the **Target** column to review the server details and status.

----End

4.4 Checking Server Details

After the Agent is installed and started on a source server, it automatically reports the source server information to SMS. The information is used for migration only. To learn what source server information is collected by SMS, see **What Information Does SMS Collect About Source Servers?** You can sign in to the SMS console to view the server information at any time. You can see source server details, target server configurations, migration status, and error messages if any.

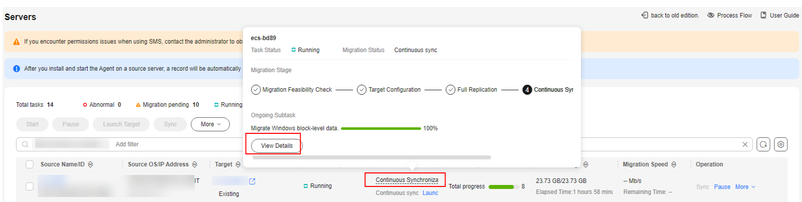
Checking Server Details

Step 1 Sign in to the **SMS console**.

Step 2 In the navigation pane, click **Servers**.

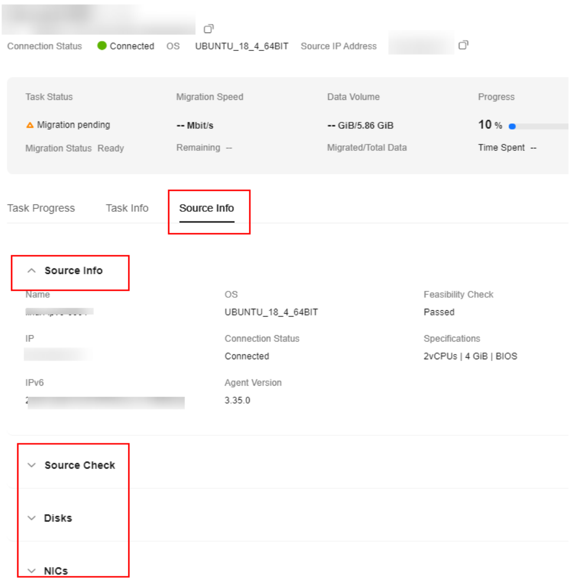
Step 3 In the server list, click the server name. The task details show up on the right.

You can also move the cursor to the migration stage and click **View Details** in the displayed window. The task details show up on the right.



Step 4 Click the **Source Info** tab, and you can check the source server details, including the basic information, migration check results, disk and partition information, and NIC information.

Figure 4-3 Checking server details



----End

4.5 Deleting the Target Server Configuration

If the target server configuration in a migration task is incorrect or needs to be updated, you can delete the configuration and reconfigure the target server.

CAUTION

- After the configuration of the target server is deleted, completed migration tasks and snapshots will be deleted. All data needs to be migrated again.
- After the target server configuration is deleted, the migration task remains in the list, but the target server settings will be gone. Incremental synchronization will no longer be possible. You can reconfigure the target server and perform a full data migration from the source server again.
- Deleting the target server configuration sometimes cannot delete the temporary SMS disks. You need to manually delete them. For details, see **How Do I Manually Detach the Temporary System Disk from My Target Server and Re-attach the Original System Disk?**

Deleting the Target Server Configuration

Step 1 Sign in to the **SMS console**.

Step 2 In the navigation pane, click **Servers**.

Step 3 Locate the server for which you want to delete the target server configuration, and choose **More** > **Delete Target Configuration** in the **Operation** column.

Alternatively, select the server for which you want to delete the target server configuration, and choose **More** > **Delete Target Configuration** above the server list.

Step 4 Read the precautions carefully in the **Delete Target Configuration** dialog box. After confirming the deletion, enter **DELETE** in the text box and click **OK**.

In the server list, if the task status changes to **Pending target configuration**, the target configuration is deleted successfully.

----End

4.6 (Optional) Deleting a Server Clone

You can delete a server clone when it is no longer needed or the service tests are complete.

Deleting a Clone Server

Step 1 Sign in to the **SMS console**.

Step 2 In the navigation pane, click **Servers**.

Step 3 Locate the server for which you want to delete the clone, and choose **More** > **Manage Target** > **Delete Clone** in the **Operation** column.

Step 4 In the **Delete Clone** dialog box, click **OK**. After deleting the server clone, verify the deletion on the **ECS console**. The deletion is successful when the server clone no longer appears in the ECS list.

----End

5 Template Management

5.1 Overview

You can create two types of templates to streamline migration and target server configuration.

Table 5-1 Templates

Template Type	Description	Operation
Migration templates	You can create migration templates to streamline migration and target server configuration. The template contains parameters such as **Network Type**, **Migration Rate Limit**, **Enable Continuous Synchronization**, and **Region/ Project**. You can modify your migration templates at any time.	• **Creating a Migration Template** • **Modifying a Migration Template** • **Deleting a Migration Template**
Server templates	You can create a server template to define the environment settings for servers, such as VPC, subnet, and security group settings. You can modify your server templates at any time.	• **Creating a Server Template** • **Modifying a Server Template** • **Deleting a Server Template**

5.2 Managing a Migration Template

A migration template defines the settings for **Network**, **Migration Rate Limit**, **Enable Continuous Synchronization**, **Region/Project**, and other migration parameters. When migrating multiple servers, you can select a template to automatically fill in parameters, improving migration efficiency.

Templates can be modified as required. Unnecessary templates can also be deleted. For details, see:

- **Creating a Migration Template**
- **Modifying a Migration Template**
- **Deleting a Migration Template**

Constraints

Custom migration parameter templates are only available on the migration target configuration page in the legacy console.

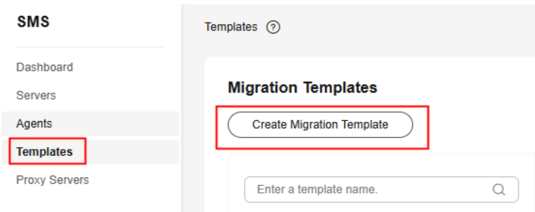
Creating a Migration Template

Step 1 Sign in to the **SMS console**.

Step 2 In the navigation pane, choose **Templates**.

Step 3 In the upper left corner of the **Migration Templates** area, click **Create Migration Template**.

Figure 5-1 Creating a migration template



Step 4 Set **Name** and **Description** and click **OK**.

Step 5 In the template list on the left of the **Migration Templates** area, click the created template and click **Parameter Settings** to configure the template, as shown in **Figure 5-2**.

Figure 5-2 Parameter settings

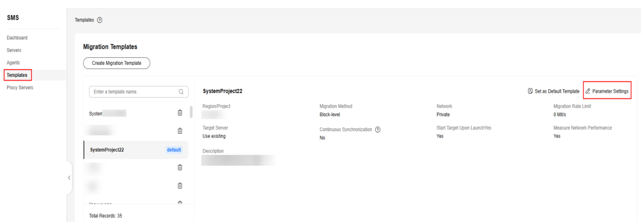


Table 5-2 describes the parameters.

Table 5-2 Parameters

Parameter	Option	Description
Name	-	User-defined
Description	-	User-defined
Region/Project	-	Select the target region and project you want to migrate to.
Migration Method	Block-level	● Migration and synchronization are performed by block. ● For Windows servers, SMS only supports block-level migration.
	File-level	Migration and synchronization are performed by file. This method is inefficient, but the compatibility is excellent.
Network	Public	Migration over the Internet requires that the target server has an EIP bound. **Public** is the default value.
	Private	You need to create a Direct Connect or VPN connection between the source and the VPC subnet you are migrating to. If the source and target servers are in the same VPC, select **Private**.
Migration Rate Limit	-	You can limit the migration rate based on the source bandwidth and service requirements. If you do not want to limit the migration rate, set this parameter to **0**.

Parameter	Option	Description
Target Server	Use existing	When you apply this template to a source server migration, you can select an existing server as the target server. The chosen server must meet at least the system-recommended specifications.
	Create new	When you apply this template to a source server migration, you need to configure environment settings for the target server, such as VPC, subnet, and security group.

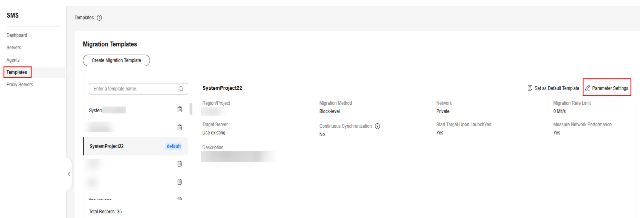
Parameter	Option	Description
Enable Continuous Synchronization	-	• If you do not enable this option, after the full replication is complete, SMS will automatically launch the target server without synchronizing incremental data. To synchronize incremental data, you will need to click **Sync** in the **Operation** column. • If you enable this option, after the full replication is complete, the migration will enter the continuous synchronization stage. During this stage, incremental data will be periodically synchronized from the source server to the target server, and you will be unable to use the target server since it has not been launched yet. To finish this stage, you will need to click **Launch Target** in the **Operation** column.
Start Target Upon Launch	-	• If you enable this option, the target server will be started after the migration is complete. • If you do not enable this option, the target server will be stopped after the migration is complete.

Parameter	Option	Description
Measure Network Performance	-	• If you enable this option, before the full migration starts, the system will measure the packet loss rate, network jitter, network latency, bandwidth, memory usage, and CPU usage for the source server. For details, see **How Do I Measure the Network Performance Before the Migration?** • If you do not enable this option, network performance will not be measured.

Step 6 Click **OK**.

Step 7 (Optional) Click the name of the created template, and click **Set as Default Template** to set it as the default template, as shown in **Figure 5-3**.

Figure 5-3 Set as Default Template



----End

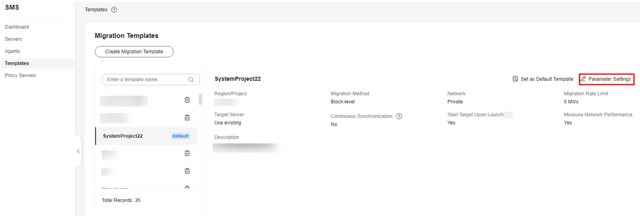
Modifying a Migration Template

Step 1 Sign in to the **SMS console**.

Step 2 In the navigation pane, choose **Templates**.

Step 3 In the template list on the left of the **Migration Templates** area, click the name of the template to be modified and click **Parameter Settings**, as shown in **Figure 5-4**.

Figure 5-4 Modifying template parameters



Step 4 Modify the template settings and click **OK**.

----End

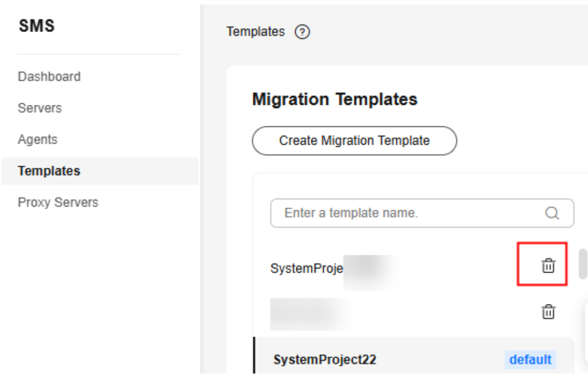
Deleting a Migration Template

Step 1 Sign in to the **SMS console**.

Step 2 In the navigation pane, choose **Templates**.

Step 3 In the **Migration Templates** area, on the left, click next to the name of the template you want to delete, as shown in **Figure 5-5**.

Figure 5-5 Deleting a migration template



Step 4 In the displayed **Delete Migration Template** dialog box, click **OK**.

----End

5.3 Managing a Server Template

A server template defines the environment settings for servers, such as VPC, subnet, and security group settings. When migrating multiple servers, you can select a template to automatically fill in parameters, improving migration efficiency.

Templates can be modified as required. Unnecessary templates can also be deleted. For details, see:

- **Creating a Server Template**
- **Modifying a Server Template**
- **Deleting a Server Template**

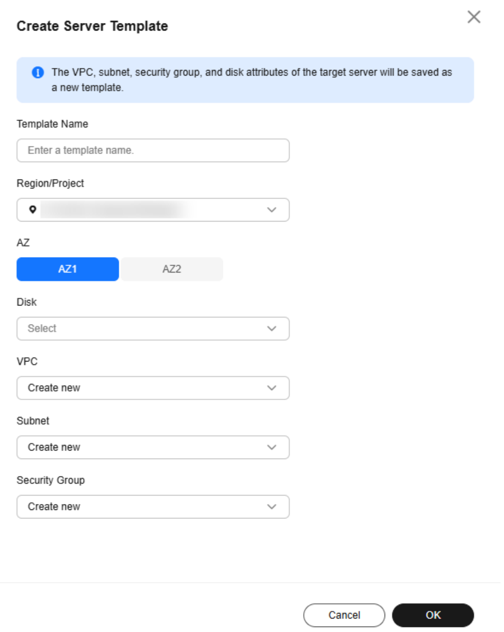
Creating a Server Template

Step 1 Sign in to the **SMS console**.

Step 2 In the navigation pane, choose **Templates**.

Step 3 In the upper right corner of the **Server Templates** area, click **Create Server Template**.

Figure 5-6 Creating a server template



Step 4 Set template parameters according to **Table 5-3**.

Table 5-3 Parameters required for creating a server template

Parameter	Description
Template Name	User-defined
Region/Project	Select a region and project where you want to provision and manage target servers. By default, the region is the one set in the default migration template, but you can change it as needed.
AZ	The parameter is set to **Random** by default. You can also select another AZ.
Disk	Select a disk type as required. Available disk types depend on the AZ.

Parameter	Description
VPC	If you select **Create new**, SMS will recommend a VPC when you use this template to configure a target server. You can also create or select a VPC based on the same rules. ● If the source server's IP address is 192.168.*X.X*, the system creates a VPC and a subnet that both belong to network range 192.168.0.0/16. ● If the source server's IP address is 172.16.*X.X*, the system creates a VPC and a subnet that both belong to network range 172.16.0.0/12. ● If the source server's IP address is 10.*X.X.X*, the system creates a VPC and a subnet that both belong to network range 10.0.0.0/8.
Subnet	If you select **Create new**, SMS will recommend a subnet when you use this template to configure a target server. The subnet must be in the same network range as the VPC.
Security Group	If you select **Create new**, SMS will recommend a security group when you use this template to configure a target server. You can also create or select a security group that opens the following inbound ports: ● Windows: ports 8899, 8900, and 22 ● Linux: port 22 for file-level migration, and ports 8900 and 22 for block-level migration **CAUTION** – For security purposes, you are advised to only allow traffic from the source server over these ports. – The firewall of the ECS must allow traffic to these ports.

Step 5 Click **OK**.

----End

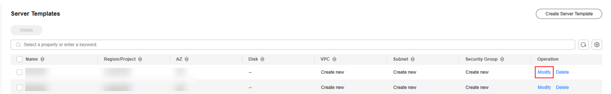
Modifying a Server Template

Step 1 Sign in to the **SMS console**.

Step 2 In the navigation pane, choose **Templates**.

Step 3 Locate the server template to be modified and click **Modify** in the **Operation** column.

Figure 5-7 Modifying a server template



Step 4 Modify the template settings and click **OK**.

----End

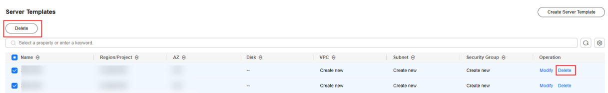
Deleting a Server Template

Step 1 Sign in to the **SMS console**.

Step 2 In the navigation pane, choose **Templates**.

Step 3 Locate the server template to be deleted and click **Delete** in the **Operation** column. If multiple templates need to be deleted, select them and click **Delete** above the list.

Figure 5-8 Deleting server templates



Step 4 Click **OK**.

----End

6 Viewing CTS Traces

6.1 SMS Operations Supported by CTS

Table 6-1 SMS operations recorded by CTS

Operation	Resource Type	Trace Name
Obtaining the Agent configuration information	config	getConfig
Querying an SMS API version	api	getApiInfo
Listing SMS API versions	api	listApi
Obtaining commands from SMS	TaskCommand	getTaskCommand
Reporting command execution results to SMS	commandResult	processCommandResult
Obtaining consistency verification results in batches	ConsistencyCheckResult	GetBatchConsistency-CheckResult
Obtaining consistency verification results	ConsistencyCheckResult	GetConsistencyCheckRe-sult
Uploading consistency verification results	ConsistencyCheckResult	UpdateConsistencyResult
Obtaining an SSL certificate and private key	CertKey	getCertKey
Creating a Migration Project	MigProject	addMigProject

Operation	Resource Type	Trace Name
Listing migration projects	MigProject	listMigProject
Querying details about a migration project with a specified ID	MigProject	getMigProject
Changing the default migration project	MigProject	setMigProjectDefault
Deleting a migration project	MigProject	removeMigProject
Modifying a migration project	MigProject	updateMigProject
Updating network measurement information	Task	updateNetworkCheckIn-fo
Agreeing to the privacy agreement	PrivacyAgreement	CreatePrivacyAgreements
Checking whether a user has agreed to the privacy agreement	PrivacyAgreement	GetPrivacyAgreement
Registering a source server with SMS	SourceServer	RegisterSourceServer
Obtaining an overview of source servers	allSourceServer	getOverview
Deleting a source server record	SourceServer	removeSource
Batch deleting source server records	SourceServer	removeSources
Updating disk information	SourceServer	updateSourceDiskInfo
Modifying information about a source server with a specified ID	SourceServer	updateSource
Querying a source server with a specified ID	SourceServer	findSourceServerById
Updating the migration status of a source server	SourceServerStatus	updateCopyState
Listing source servers	SourceServer	listSourceServers
Listing failed source servers	ErrorInform	listSourceErrorInform

Operation	Resource Type	Trace Name
Querying the settings of advanced migration options of a task	TaskConfig	getTaskConfig
Configuring advanced migration options	Task	updateSpecialConfigSetting
Querying the migration rate limiting rules of a migration task	taskSpeedLimit	getSpeedLimit
Setting migration rate limiting rules for a migration task	taskSpeed	updateSpeedLimit
Uploading migration task logs	collectLog	task-collect-log-request
Creating a migration task	Task	CreateTask
Managing a migration task	Task	updateTaskStatus
Updating a migration task with a specified ID	Task	UpdateTask
Querying the certificate passphrase of the secure transmission channel for a task	taskPassPhrase	getTaskPassPhrase
Deleting a migration task with a specified ID	Task	deleteTask
Deleting migration tasks in batches	Task	deleteTasks
Reporting the data migration progress and rate	Task	updateTaskProgressSpeed
Listing migration tasks	Task	getTasks
Querying a migration task with a specified ID	Task	getTask
Creating a template	Template	addTemplate
Deleting a template with a specified ID	Template	deleteTemplate
Listing templates	Templates	getTemplates

Operation	Resource Type	Trace Name
Querying information about a template with a specified ID	Template	getTemplate
Modifying template information	Template	updateTemplate
Deleting templates with specified IDs in batches	Template	deleteTemplates
Querying the target server password in a template with a specified ID	TemplatePassword	getTargetPassword

6.2 Viewing CTS Traces in the Trace List

Scenarios

After you enable CTS and the management tracker is created, CTS starts recording operations on cloud resources. After a data tracker is created, the system starts recording operations on data in Object Storage Service (OBS) buckets. Cloud Trace Service (CTS) stores operation records (traces) generated in the last seven days.

This section describes how to query or export operation records of the last seven days on the CTS console.

- **Viewing Real-Time Traces in the Trace List of the New Edition**
- **Viewing Real-Time Traces in the Trace List of the Old Edition**

Constraints

- Traces of a single account can be viewed on the CTS console. Multi-account traces can be viewed only on the **Trace List** page of each account, or in the OBS bucket or the **CTS/system** log stream configured for the management tracker with the organization function enabled.
- You can only query operation records of the last seven days on the CTS console. To store operation records for longer than seven days, you must configure transfer to OBS or Log Tank Service (LTS) so that you can view them in OBS buckets or LTS log groups.
- After performing operations on the cloud, you can query management traces on the CTS console one minute later and query data traces five minutes later.
- These operation records are retained for seven days on the CTS console and are automatically deleted upon expiration. Manual deletion is not supported.

Viewing Real-Time Traces in the Trace List of the New Edition

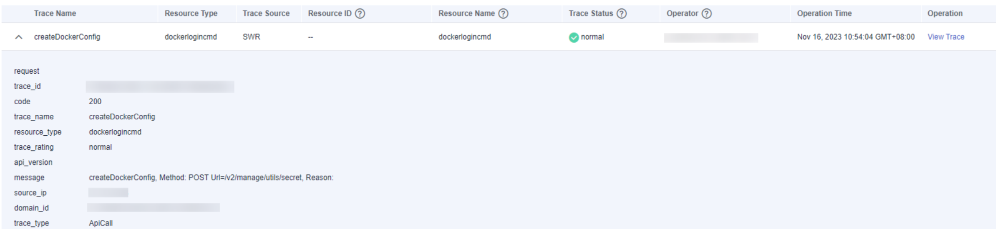
1. Log in to the **CTS console**.

2. In the navigation pane on the left, choose **Trace List**.
3. On the **Trace List** page, use advanced search to query traces. You can combine one or more filters.
   - **Trace Name**: Enter a trace name.
   - **Trace ID**: Enter a trace ID.
   - **Resource Name**: Enter a resource name. If the cloud resource involved in the trace does not have a resource name or the corresponding API operation does not involve the resource name parameter, leave this field empty.
   - **Resource ID**: Enter a resource ID. Leave this field empty if the resource has no resource ID or if resource creation failed.
   - **Trace Source**: Select a cloud service name from the drop-down list.
   - **Resource Type**: Select a resource type from the drop-down list.
   - **Operator**: Select one or more operators from the drop-down list.
   - **Trace Status**: Select **normal**, **warning**, or **incident**.
     - **normal**: The operation succeeded.
     - **warning**: The operation failed.
     - **incident**: The operation caused a fault that is more serious than the operation failure, for example, causing other faults.
   - **Enterprise Project ID**: Enter an enterprise project ID.
   - **Access Key**: Enter a temporary or permanent access key ID.
   - Time range: Select **Last 1 hour**, **Last 1 day**, or **Last 1 week**, or specify a custom time range within the last seven days.
4. On the **Trace List** page, you can also export and refresh the trace list, and customize columns to display.
   - Enter any keyword in the search box and press **Enter** to filter desired traces.
   - Click **Export** to export all traces in the query result as an .xlsx file. The file can contain up to 5,000 records.
   - Click to view the latest information about traces.
   - Click to customize the information to be displayed in the trace list. If **Auto wrapping** is enabled ( ), excess text will move down to the next line; otherwise, the text will be truncated. By default, this function is disabled.
5. For details about key fields in the trace structure, see **Trace Structure** and **Example Traces**.
6. (Optional) On the **Trace List** page of the new edition, click **Go to Old Edition** in the upper right corner to switch to the **Trace List** page of the old edition.

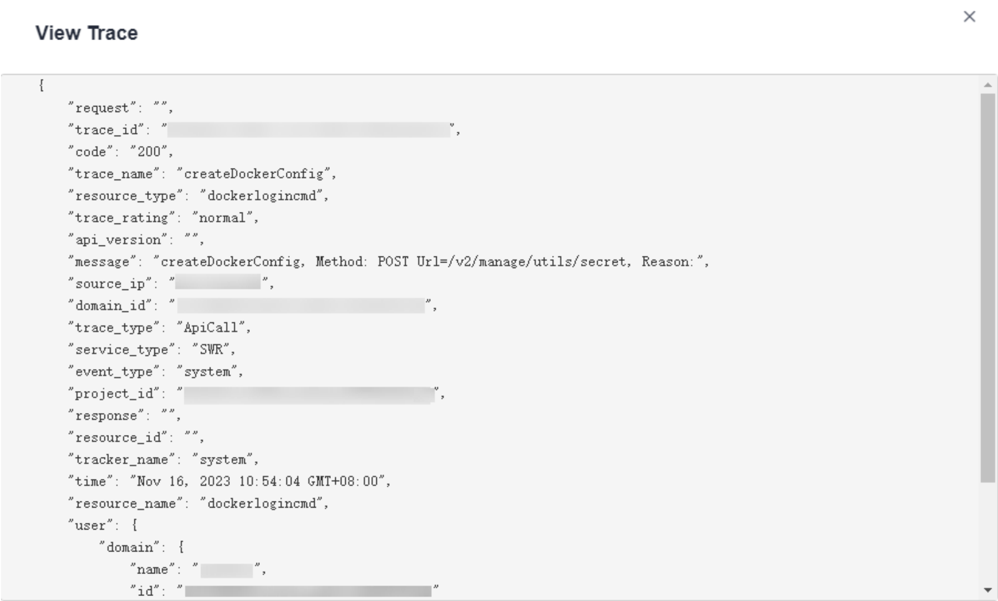
Viewing Real-Time Traces in the Trace List of the Old Edition

1. Log in to the **CTS console**.
2. In the navigation pane on the left, choose **Trace List**.

3. Each time you log in to the CTS console, the new edition is displayed by default. Click **Go to Old Edition** in the upper right corner to switch to the trace list of the old edition.
4. Set filters to search for your desired traces. The filters below are available.
   - **Trace Type**, **Trace Source**, **Resource Type**, and **Search By**: Select a filter from the drop-down list.
     - If you select **Resource ID** for **Search By**, specify a resource ID.
     - If you select **Trace name** for **Search By**, specify a trace name.
     - If you select **Resource name** for **Search By**, specify a resource name.
   - **Operator**: Select a user.
   - **Trace Status**: Select **All trace statuses**, **Normal**, **Warning**, or **Incident**.
   - Time range: Select **Last 1 hour**, **Last 1 day**, or **Last 1 week**, or specify a custom time range within the last seven days.
5. Click **Query**.
6. On the **Trace List** page, you can also export and refresh the trace list.
   - Click **Export** to export all traces in the query result as a CSV file. The file can contain up to 5,000 records.
   - Click to view the latest information about traces.
7. Click on the left of a trace to expand its details.



8. Click **View Trace** in the **Operation** column. The trace details are displayed.



9. For details about key fields in the trace structure, see **Trace Structure** and **Example Traces** in the *CTS User Guide*.
10. (Optional) On the **Trace List** page of the old edition, click **New Edition** in the upper right corner to switch to the **Trace List** page of the new edition.

6.3 Viewing Traces

Scenarios

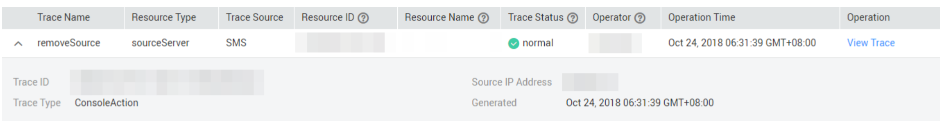
After you enable CTS, it records key operations performed on SMS. You can view the operation records of the last seven days on the CTS console.

Procedure

1. Log in to the **CTS console**.
2. In the navigation pane on the left, choose **Trace List**.
3. In the upper right corner of the trace list, click **Filter** to set the search criteria. The filters below are available.
   - **Trace Type**, **Trace Source**, **Resource Type**, and **Search By**: Select a filter from the drop-down list.

     If you select **Resource ID** for **Search By**, specify a resource ID.
   - **Operator**: Select a specific operator from the drop-down list.
   - **Trace Status**: Select **All trace statuses**, **Normal**, **Warning**, or **Incident**.
   - Time range: In the upper right corner of the page, you can query traces in the last one hour, last one day, last one week, or within a customized period.
4. Click **Query**.
5. On the right of the filter box, click **Export**. CTS exports a CSV file which lists query results.
6. Click on the left of the required trace to expand its details. **Figure 6-1** shows an example.

Figure 6-1 Expanding trace details



7. Click **View Trace** in the **Operation** column. The trace structure details are displayed.

